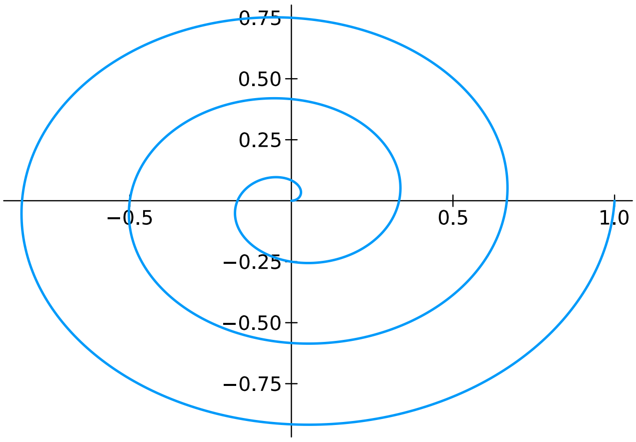
11 Analysis in Coordinates

We begin with a coordinate-free definition and then introduce suitable coordinates that allow for explicit computations. Curves provide a natural illustration of this principle: although defined abstractly in metric spaces, their length can be computed in coordinates by integrating the norm of the derivative.

Curves

Let X be a metric space. A map $f : [0,1] \to X$ is also called a curve. If $X = \mathbb{R}^n$, then f is called a space curve.

```
function myCurvePlot(f;n=1000)
    f1 = t->f(t)[1]
    f2 = t->f(t)[2]
    t = range(0,1,n)
    plot(f1.(t),f2.(t),lw=2)
end

f(t) = [t*cos(6*π*t), t*sin(6*π*t)]
myCurvePlot(f)
```



Definition 11.0.1 (Length of a curve in a metric space). *Let (X,d) be a metric space and let $\gamma : [0,1] \to X$ be a curve. For a partition*

$$\mathcal{P} = \{0 = t_0 < t_1 < \cdots < t_n = 1\}$$

of $[0,1]$, define the polygonal length of γ along $\mathcal{P}$ by

$$\ell(\gamma,\mathcal{P}) := \sum_{j=1}^{n} d\big(\gamma(t_{j-1}),\gamma(t_j)\big).$$

The length *of* γ *is*

$$\ell(\gamma) := \sup_{\mathcal{P}} \ell(\gamma, \mathcal{P}),$$

where the supremum is taken over all partitions $\mathcal{P}$ *of* $[0,1]$*. The curve* γ *is called* rectifiable *if* $\ell(\gamma) < \infty$*.*

Definition 11.0.2. *Let* (X, d) *be a metric space and let* $\gamma : [0,1] \to X$ *be a curve. If the limit exists,*

$$|\dot{\gamma}|(t) = \lim_{h \to 0} \frac{d(\gamma(t+h), \gamma(t))}{|h|},$$

then $|\dot{\gamma}|(t)$ *is called the metric derivative of* γ *in* t*.*

For $X = \mathbb{R}^n$, we choose the standard metric

$$d(x, y) = \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2} = \|x - y\|_2.$$

A function $f : \mathbb{R} \to \mathbb{R}^n$ is written as

$$f(t) = \begin{pmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{pmatrix}.$$

We call f differentiable in t_0 if each $f_i : \mathbb{R} \to \mathbb{R}$, for $i = 1, \ldots, n$, is differentiable in t_0 . In that case we write

$$f'(t_0) = \begin{pmatrix} f_1'(t_0) \\ \vdots \\ f_n'(t_0) \end{pmatrix}.$$

This makes sense because the linear map

$$h \mapsto f'(t_0)h \in \mathbb{R}^n$$

approximates f locally in the sense

$$f(t_0 + h) = f(t_0) + f'(t_0)h + \varphi(h),$$

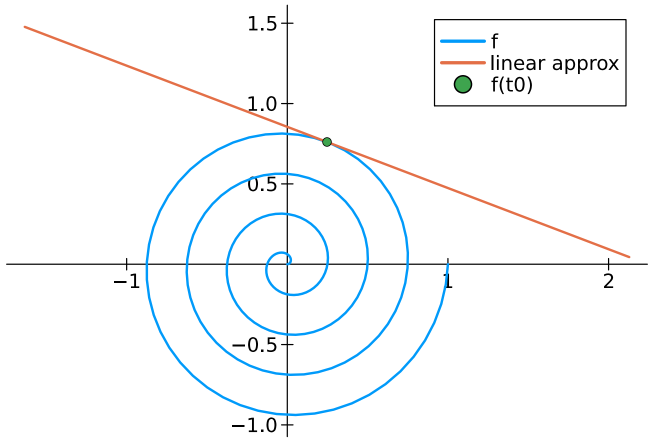
where $\varphi(h)/h \to 0$ for $h \to 0$. This is equivalent to

$$f_i(t_0 + h) = f_i(t_0) + f_i'(t_0)h + \varphi_i(h), \qquad i = 1, \ldots, n,$$

where $\varphi_i(h)/h \to 0$ for $h \to 0$.

```julia
using Calculus
f(t) = [t*cos(8*π*t), t*sin(8*π*t)]

function myAffineApprox(f, t0)
    n = length(f(0))
    df0 = [derivative(h -> f(h)[j], t0) for j in 1:n]
    return h -> f(t0) + h * df0
end

plot(t -> f(t)[1], t -> f(t)[2], 0, 1, label="f", lw=2, aspect_ratio=1)
```

```
t0 = .8
A = myAffineApprox(f, t0)
plot!(t -> A(t)[1], t -> A(t)[2], -0.1, 0.1, label="linear approx", lw=2)
scatter!([f(t0)[1]], [f(t0)[2]], label="f(t0)")
```



Definition 11.0.3. *Let (X, d) be a metric space and let $\gamma : [0,1] \to X$ be a curve. We say that γ is Lipschitz continuous if there is a constant $L > 0$ such that*

$$d(\gamma(t), \gamma(s)) \leq L|t - s|, \qquad t, s \in [0,1].$$

Lemma 11.0.4. *If the curve $\gamma : [0,1] \to \mathbb{R}^n$ is continuously differentiable, then it is Lipschitz continuous and its metric derivative $|\dot{\gamma}|$ satisfies*

$$|\dot{\gamma}|(t) = \|\gamma'(t)\|_2.$$

In particular, $|\dot{\gamma}|$ is continuous on $[0,1]$.

Proof. For $h \neq 0$, we calculate

$$\begin{aligned}\frac{\|\gamma(t+h) - \gamma(t)\|_2}{|h|} &= \left(\sum_{i=1}^{n} \frac{|\gamma_i(t+h) - \gamma_i(t)|^2}{|h|^2}\right)^{1/2} \\ &= \left(\sum_{i=1}^{n} \left|\frac{\gamma_i(t+h) - \gamma_i(t)}{h}\right|^2\right)^{1/2} \to \left(\sum_{i=1}^{n} |\gamma_i'(t)|^2\right)^{1/2} = \|\gamma'(t)\|_2,\end{aligned}$$

where $h \to 0$. Thus, metric derivative satisfies $|\dot{\gamma}|(t) = \|\gamma'(t)\|_2$.
To verify Lipschitz continuity, we first recall that $[0,1]$ is compact, hence $\gamma_i'([0,1]) \subseteq \mathbb{R}$ is compact since γ_i' is continuous. Taking the maximum over $i = 1, \ldots, n$, there is $C > 0$ such that

$$|\gamma_i'(t)| \leq C, \qquad t \in [0,1], \quad i = 1, \ldots, n.$$

For $s, t \in [0,1]$ the mean value theorem ensures that

$$|\gamma_i(s) - \gamma_i(t)| \leq C|s - t|.$$

Since $d_2 \leq \sqrt{n} d_\infty$, we deduce

$$\|\gamma(s) - \gamma(t)\|_2 \leq \underbrace{\sqrt{n}C}_{L} |s - t|.$$

$\square$

Theorem 11.0.5. *Let $\gamma : [0,1] \to \mathbb{R}^d$ be continuously differentiable. Then*

$$\ell(\gamma) = \int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_d'(t)|^2}\, dt = \int_0^1 \|\gamma'(t)\|_2 \, dt.$$

Proof. The function $g(t) \mapsto \|\gamma'(t)\|_2$ is continuous on $[0,1]$.
Step 1: $\ell(\gamma) \leq \int_0^1 g(t)\, dt$. Let $\mathcal{P} = \{t_0, \ldots, t_m\}$ be a partition of $[0,1]$. For each j and each component, the mean value theorem gives

$$\gamma_k(t_j) - \gamma_k(t_{j-1}) = \gamma_k'(\xi_{j,k})(t_j - t_{j-1})$$

for some $\xi_{j,k} \in (t_{j-1}, t_j)$. Hence

$$|\gamma_k(t_j) - \gamma_k(t_{j-1})| \leq \sup_{t \in [t_{j-1}, t_j]} |\gamma_k'(t)| \, (t_j - t_{j-1}).$$

Therefore

$$\sqrt{\sum_{k=1}^n |\gamma_k(t_j) - \gamma_k(t_{j-1})|^2} \leq \sup_{t \in [t_{j-1}, t_j]} g(t) \, (t_j - t_{j-1}).$$

Summing over j and taking the supremum over all partitions yields

$$\ell(\gamma) \leq \int_0^1 \|\gamma'(t)\|_2 \, dt.$$

Step 2: $\ell(\gamma) \geq \int_0^1 g(t)\, dt$. Since g is continuous, it is uniformly continuous. Let $\varepsilon > 0$ and choose a partition $\mathcal{P}$ with mesh sufficiently small so that

$$\sqrt{\sum_{k=1}^n |\gamma_k'(s) - \gamma_k'(t)|^2} < \varepsilon \quad \text{whenever } s, t \text{ lie in one subinterval.}$$

For each j, pick $\tau_j \in [t_{j-1}, t_j]$. Using the mean value theorem componentwise as above, we obtain

$$\left| \frac{\gamma_k(t_j) - \gamma_k(t_{j-1})}{t_j - t_{j-1}} - \gamma_k'(\tau_j) \right| \leq \varepsilon \qquad (k = 1, \ldots, n).$$

Hence

$$\sqrt{\sum_{k=1}^n \left| \frac{\gamma_k(t_j) - \gamma_k(t_{j-1})}{t_j - t_{j-1}} - \gamma_k'(\tau_j) \right|^2} \leq \sqrt{n}\, \varepsilon.$$

Thus

$$\sqrt{\sum_{k=1}^n |\gamma_k(t_j) - \gamma_k(t_{j-1})|^2} \geq \big(g(\tau_j) - \sqrt{n}\, \varepsilon\big)(t_j - t_{j-1}).$$

Summing gives

$$\ell(\gamma, \mathcal{P}) \geq \sum_{j=1}^{m} g(\tau_j)(t_j - t_{j-1}) - \sqrt{n}\,\varepsilon.$$

Passing to the limit (Riemann sums) and using $\ell(\gamma) \geq \ell(\gamma, \mathcal{P})$,

$$\ell(\gamma) \geq \int_0^1 g(t)\,dt.$$

Combining both inequalities proves

$$\ell(\gamma) = \int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_n'(t)|^2}\,dt.$$

□

Example 11.0.6 (circumference of the unit circle)**.** *The curve $\gamma : [0,1] \to \mathbb{R}^2$ given by*

$$\gamma(t) = \begin{pmatrix} \sin(2\pi t) \\ \cos(2\pi t) \end{pmatrix}$$

is a circle. Its length is

$$\begin{aligned} \ell(\gamma) &= \int_0^1 \sqrt{|2\pi\cos(2\pi t)|^2 + |-2\pi\sin(2\pi t)|^2}\,dt \\ &= 2\pi \int_0^1 \sqrt{\cos^2(2\pi t) + \sin^2(2\pi t)}\,dt \\ &= 2\pi \int_0^1 dt = 2\pi. \end{aligned}$$

```
function myPolygonLength(f;n=1000)
    t = range(0, 1, n)
    sum(norm(f(t[j+1]) - f(t[j])) for j in 1:n-1)
end

g(t) = [sin(2*π*t);cos(2*π*t)]
myPolygonLength(g), 2*π
```

```
(6.283174951057152, 6.283185307179586)
```

The length of a curve does not depend on the parametrization.

Lemma 11.0.7. *Let $f : [a,b] \to \mathbb{R}^n$ be a continuously differentiable function and let $\phi : [0,1] \to [a,b]$ be bijective and continuously differentiable. Then $\gamma := f \circ \phi : [0,1] \to \mathbb{R}^n$ satisfies*

$$\int_a^b \sqrt{|f_1'(x)|^2 + \cdots + |f_n'(x)|^2}\,dx = \int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_n'(t)|^2}\,dt.$$

Proof. The function ϕ is either strictly increasing or decreasing. Assume it is increasing, so that $\phi'(t) \geq 0$ with $\phi(0) = a$ and $\phi(1) = b$. Substituting $x = \phi(t)$ yields

$$\begin{aligned}\int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_n'(t)|^2}\, dt &= \int_0^1 \sqrt{|f_1'(\phi(t))\phi'(t)|^2 + \cdots + |f_n'(\phi(t))\phi'(t)|^2}\, dt \\ &= \int_0^1 \sqrt{|f_1'(\phi(t))|^2 + \cdots + |f_n'(\phi(t))|^2}\, \phi'(t)\, dt \\ &= \int_a^b \sqrt{|f_1'(x)|^2 + \cdots + |f_n'(x)|^2}\, dx.\end{aligned}$$

The case that ϕ is decreasing is analogous. $\square$

11.1 Differentiability in coordinates

We begin in an infinite-dimensional, coordinate-free setting and then introduce a suitable representation together with a finite-dimensional truncation. This reduces the problem from the Hilbert space H to the finite-dimensional coordinate space $\mathbb{R}^N$, where explicit computations become possible.

Given a sequence $(\phi_n)_{n\in\mathbb{N}} \subseteq H$, we define the analysis (decomposition) operator by

$$\mathcal{D}_\phi x := (\langle x, \phi_n\rangle)_{n\in\mathbb{N}}.$$

If we suppose that $\mathcal{D}_\phi : H \to \ell^2(\mathbb{N})$ is bounded, then its adjoint operator is

$$\mathcal{D}_\phi^* : \ell^2(\mathbb{N}) \to H, \qquad \mathcal{D}_\phi^* c = \sum_{n=1}^{\infty} c_n \phi_n.$$

Definition 11.1.1. *Let H be a Hilbert space. We call $(\phi_n)_{n\in\mathbb{N}}, (\psi_n)_{n\in\mathbb{N}} \subseteq H$ a dual frame pair if both $\mathcal{D}_\phi, \mathcal{D}_\psi : H \to \ell^2(\mathbb{N})$ are bounded linear operators such that*

$$\mathcal{D}_\psi^* \mathcal{D}_\phi = \mathrm{id}_H\,.$$

Thus, a dual frame pair satisfies the reconstruction formula

$$\sum_{n\in\mathbb{N}} \langle x, \phi_n\rangle \psi_n = x, \qquad \forall\, x \in H.$$

Example 11.1.2. *Suppose that $(\phi_n)_{n\in\mathbb{N}} \subseteq H$ is an orthonormal basis. Then $\mathcal{D}_\phi : H \to \ell^2(\mathbb{N})$ is bounded and the expansion*

$$x = \sum_{n\in\mathbb{N}} \langle x, \phi_n\rangle \phi_n, \qquad \forall\, x \in H,$$

is exactly the dual frame pair condition $\mathcal{D}_\phi^* \mathcal{D}_\phi = \mathrm{id}_H$. Hence, every orthonormal basis with itself builds a pair of dual frames.

Dual frame pairs generalize the concept of orthonormal basis.

Let truncation and its adjoint be denoted by

$$\mathcal{T}_N : \ell^2(\mathbb{N}) \to \mathbb{R}^N, \qquad \mathcal{T}_N(c_1, c_2, \dots) = (c_1, \dots, c_N)^\top,$$
$$\mathcal{T}_N^* : \mathbb{R}^N \to \ell^2(\mathbb{N}), \qquad \mathcal{T}_N^*(c_1, \dots, c_N)^\top = (c_1, \dots, c_N, 0, \dots).$$

Define the truncated reconstruction maps on H by

$$Q_N : H \to \operatorname{span}\{\psi_n : n \leq N\}, \qquad Q_N = \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi.$$

The explicit representation is

$$Q_N x = \sum_{n=1}^{N} \langle x, \phi_n \rangle \psi_n.$$

Lemma 11.1.3. *Let H be a Hilbert space and let $(\phi_n)_{n\in\mathbb{N}}$ and $(\psi_n)_{n\in\mathbb{N}}$ be a pair of dual frames. Then Q_N is uniformly bounded and converges pointwise to the identity,*

$$\|Q_N\| \leq \|\mathcal{D}_\psi\| \, \|\mathcal{D}_\phi\|, \qquad Q_N \to \mathrm{id}_H, \quad \textit{pointwise.}$$

Proof. The pointwise convergence is just the dual frame expansion. The norm bound follows from $\|\mathcal{T}_N^* \circ \mathcal{T}_N\| \leq 1$. □

Theorem 11.1.4 (R T A Representation, truncation for continuous maps I)**.** *Let H be a Hilbert space and let $(\phi_n)_{n\in\mathbb{N}}$ and $(\psi_n)_{n\in\mathbb{N}}$ be a pair of dual frames. Suppose that $f : H \to H$ is continuous. Define its representation by*

$$\tilde{f} : \ell^2(\mathbb{N}) \to \ell^2(\mathbb{N}), \qquad \tilde{f} := \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^*,$$
$$\tilde{f}_N : \mathbb{R}^N \longrightarrow \mathbb{R}^N, \qquad \tilde{f}_N := \mathcal{T}_N \circ \tilde{f} \circ \mathcal{T}_N^*.$$

We define the truncated approximation of f on H by using the truncated representation,

$$f_N : H \to \operatorname{span}\{\psi_n : n \leq N\}, \qquad f_N := \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \tilde{f}_N \circ \mathcal{T}_N \circ \mathcal{D}_\phi.$$

Then the following hold.

(i) Coordinate-free vs. coordinates:

$$f = \mathcal{D}_\psi^* \circ \tilde{f} \circ \mathcal{D}_\phi, \qquad \mathcal{D}_\phi \circ f = \tilde{f} \circ \mathcal{D}_\phi, \qquad f \circ \mathcal{D}_\psi^* = \mathcal{D}_\psi^* \circ \tilde{f}.$$

(ii) Truncated approximation: For every $N \in \mathbb{N}$,

$$f_N = Q_N \circ f \circ Q_N.$$

(iii) Coordinate-free approximation:

$$f_N \to f \quad \textit{pointwise for } N \to \infty.$$

(iv) Approximation in coordinates:

$$\mathcal{T}_N^* \circ \tilde{f}_N \circ \mathcal{T}_N \to \tilde{f} \quad \textit{pointwise for } N \to \infty.$$

We may consider f_N as a finite-dimensional approximation of f.

Proof. (i) Since (ϕ_n) and (ψ_n) are dual, $\mathcal{D}_\psi^* \mathcal{D}_\phi = \mathrm{id}_H$. Thus

$$\mathcal{D}_\psi^* \circ \tilde{f} \circ \mathcal{D}_\phi = \mathcal{D}_\psi^* \circ \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* \circ \mathcal{D}_\phi = f,$$

which gives the first identity. The other two follow from

$$\begin{aligned} \tilde{f} \circ \mathcal{D}_\phi &= \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* \circ \mathcal{D}_\phi = \mathcal{D}_\phi \circ f, \\ \mathcal{D}_\psi^* \circ \tilde{f} &= \mathcal{D}_\psi^* \circ \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* = f \circ \mathcal{D}_\psi^*. \end{aligned}$$

(ii) Using $\tilde{f} = \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^*$ and $Q_N = \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi$, we compute

$$\begin{aligned} f_N &= \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ (\mathcal{T}_N \circ \tilde{f} \circ \mathcal{T}_N^*) \circ \mathcal{T}_N \circ \mathcal{D}_\phi \\ &= \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi \\ &= Q_N \circ f \circ Q_N. \end{aligned}$$

(iii) Fix $c \in H$ and estimate

$$\begin{aligned} \|Q_N \circ f \circ Q_N(c) - f(c)\| &\le \|Q_N \circ f \circ Q_N(c) - Q_N \circ f(c)\| + \|Q_N \circ f(c) - f(c)\| \\ &\le \|Q_N\| \, \|f \circ Q_N(c) - f(c)\| + \|Q_N \circ f(c) - f(c)\|. \end{aligned}$$

Since Q_N is uniformly bounded, $Q_N c \to c$ and f is continuous, both terms tend to zero for $N \to \infty$.
(iv) Let us now denote $P_N := \mathcal{T}_N^* \circ \mathcal{T}_N$, which simply keeps the first N coordinates. According to

$$\mathcal{T}_N^* \circ \tilde{f}_N \circ \mathcal{T}_N = P_N \circ \tilde{f} \circ P_N,$$

we now estimate

$$\begin{aligned} \|P_N \circ \tilde{f} \circ P_N\,(c) - \tilde{f}(c)\| &\le \|P_N \circ \tilde{f} \circ P_N\,(c) - P_N \circ \tilde{f}(c)\| + \|P_N \circ \tilde{f}(c) - \tilde{f}(c)\| \\ &\le \|\tilde{f} \circ \mathcal{T}_N^* \circ \mathcal{T}_N\,(c) - \tilde{f}(c)\| + \|P_N \circ \tilde{f}(c) - \tilde{f}(c)\| \end{aligned}$$

Since $P_N \to \mathrm{id}_{\ell^2(\mathbb{N})}$ pointwise and $\tilde{f}$ is continuous, both terms tend to zero. □

If we allow for two Hilbert spaces, $f : H_1 \to H_2$, then we need two pairs of dual frames and truncated reconstruction operators on H_i by

$$Q_{i,N_i} : H_i \to \operatorname{span}\{\psi_{i,n} : n \le N_i\}, \qquad Q_{i,N_i} := \mathcal{D}_{\psi_i}^* \circ \mathcal{T}_{N_i}^* \circ \mathcal{T}_{N_i} \circ \mathcal{D}_{\phi_i}, \qquad Q_{i,N_i} x = \sum_{n=1}^{N_i} \langle x, \phi_{i,n} \rangle \psi_{i,n}.$$

Theorem 11.1.5 (R T A Representation, truncation for continuous maps II)**.** *Let H_1 and H_2 be Hilbert spaces and suppose that $f : H_1 \to H_2$ is continuous. Let $(\phi_{i,n})_{n\in\mathbb{N}}$ and $(\psi_{i,n})_{n\in\mathbb{N}}$ be a pair of dual frames for the Hilbert space H_i, for $i = 1, 2$. Define its representation by*

$$\tilde{f} : \ell^2(\mathbb{N}) \to \ell^2(\mathbb{N}), \qquad \tilde{f} := \mathcal{D}_{\phi_2} \circ f \circ \mathcal{D}_{\psi_1}^*,$$
$$\tilde{f}_{N_1,N_2} : \mathbb{R}^{N_1} \longrightarrow \mathbb{R}^{N_2}, \qquad \tilde{f}_{N_1,N_2} := \mathcal{T}_{N_2} \circ \tilde{f} \circ \mathcal{T}_{N_1}^*.$$

We define the truncated approximation of f on H by using the truncated representation,

$$f_{N_1,N_2} : H \to \operatorname{span}\{\psi_n : n \leq N_2\}, \qquad f_{N_1,N_2} := \mathcal{D}^*_{\psi_2} \circ \mathcal{T}^*_{N_2} \circ \tilde{f}_{N_1,N_2} \circ \mathcal{T}_{N_1} \circ \mathcal{D}_{\phi_1}.$$

Then the following hold.

(i) Coordinate-free vs. coordinates:

$$f = \mathcal{D}^*_{\psi_2} \circ \tilde{f} \circ \mathcal{D}_{\phi_1}, \qquad \mathcal{D}_{\phi_2} \circ f = \tilde{f} \circ \mathcal{D}_{\phi_1}, \qquad f \circ \mathcal{D}^*_{\psi_1} = \mathcal{D}^*_{\psi_2} \circ \tilde{f}.$$

(ii) Truncated approximation: For every $N \in \mathbb{N}$,

$$f_{N_1,N_2} = Q_{2,N_2} \circ f \circ Q_{1,N_1}.$$

(iii) Coordinate-free approximation:

$$f_{N_1,N_2} \to f \quad \text{pointwise for } N_1, N_2 \to \infty.$$

(iv) Approximation in coordinates:

$$\mathcal{T}^*_{N_2} \circ \tilde{f}_{N_1,N_2} \circ \mathcal{T}_{N_1} \to \tilde{f} \quad \text{pointwise for } N_1, N_2 \to \infty.$$

Proof. This is analogous to the proof of Theorem 11.1.4, hence omitted. □

The representation and truncation are compatible with the Fréchet derivative, so that

$$\begin{aligned} \tilde{f}' &= \mathcal{D}_{\phi_2} \circ f' \circ \mathcal{D}^*_{\psi_1}, \\ \tilde{f}'_{N_1,N_2} &= \mathcal{T}_{N_2} \circ \tilde{f}' \circ \mathcal{T}^*_{N_1}. \end{aligned}$$

Remark 11.1.6. *This representation-truncation procedure appears throughout analysis under different names. In numerical analysis it corresponds to Galerkin or Petrov-Galerkin projection methods; in harmonic analysis to matrix representations of operators with respect to frames; and in operator theory to the finite section method.*

For finite-dimensional vector spaces, we directly switch to coordinates without truncation.

Theorem 11.1.7 (reduction to coordinates)**.** *Let* $\dim V = n$, $\dim W = m$, *and* $f : V \to W$. *Choose bases* $B_V \subseteq V$, $B_W \subseteq W$ *and define* $F : \mathbb{R}^n \to \mathbb{R}^m$ *by*

$$F := \chi^{-1}_{B_W} \circ f \circ \chi_{B_V}, \qquad \text{or equivalently by} \quad f = \chi_{B_W} \circ F \circ \chi^{-1}_{B_V}.$$

Then f is Fréchet differentiable if and only if F is Fréchet differentiable. In that case, for $x = \chi^{-1}_{B_V} v$, *we have*

$$f'(v) = \chi_{B_W}\, F'(x)\, \chi^{-1}_{B_V}.$$

In contrast to linear algebra, we do not restrict to linear maps: if f is nonlinear, then its coordinate representation $F : \mathbb{R}^n \to \mathbb{R}^m$ is nonlinear as well. Fréchet differentiable maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ are often simply called *differentiable.* Thus, by fixing bases, the analysis of f can be carried out entirely on the level of its coordinate representation $F : \mathbb{R}^n \to \mathbb{R}^m$. To study differentiability, the separation

$$F = \begin{pmatrix} F_1 \\ \vdots \\ F_m \end{pmatrix}$$

allows us to first restrict us to a single $F_j : \mathbb{R}^n \to \mathbb{R}$.

We now use directional derivatives, but restrict the directions $u \in \mathbb{R}^n$ to the standard basis vectors $e_1, \dots, e_n$.

Definition 11.1.8 (Partial derivatives)**.** *The function $f : \mathbb{R}^n \to \mathbb{R}$ is called* partially differentiable *at $x \in \mathbb{R}^n$ if, for each $j = 1, \dots, n$, the limit*

$$\partial_j f(x) := \lim_{h \to 0} \frac{f(x + he_j) - f(x)}{h}$$

exists. The row vector

$$\operatorname{grad} f(x) := (\partial_1 f(x), \dots, \partial_n f(x))$$

is called the gradient *of f at x. Its transpose is a column vector and denoted by*

$$\nabla f(x) = \operatorname{grad} f(x)^\top.$$

The partial derivative $\partial_j f(x)$ is simply the derivative of the one-dimensional function

$$t \mapsto f(x_1, \dots, x_{j-1}, t, x_{j+1}, \dots, x_n)$$

at $t = x_j$, where all other coordinates are kept fixed.

Remark 11.1.9 (Feature sensitivity and gradient descent)**.** *Partial derivatives quantify how changes in individual coordinates affect the output of a multivariate model. The gradient describes the direction of strongest local change and underlies gradient-based optimization methods.*

Example 11.1.10. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x) = x_1^2 x_2 + x_2.$$

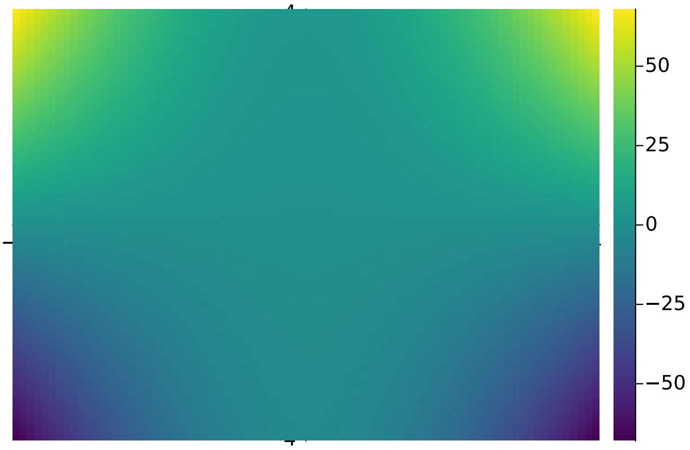
Its gradient is

$$\operatorname{grad} f(x) = \big(2x_1 x_2,\ x_1^2 + 1\big).$$

As a row it represents a linear map from $\mathbb{R}^2$ to $\mathbb{R}$.
We now check the local approximation:

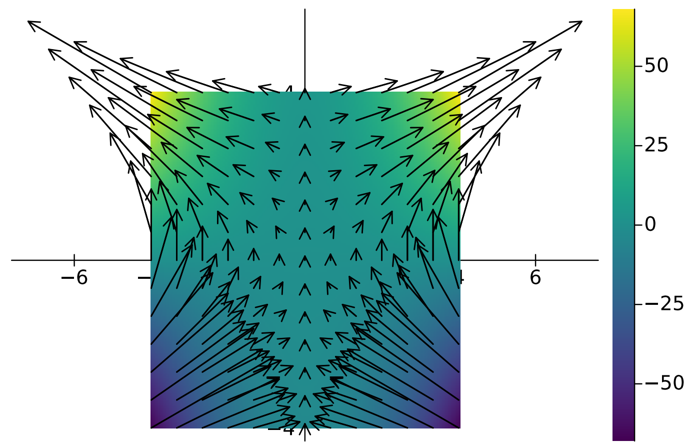
$$\begin{aligned} f(x+h) &= (x_1 + h_1)^2 (x_2 + h_2) + x_2 + h_2 \\ &= x_1^2 x_2 + x_1^2 h_2 + 2x_1 h_1 x_2 + 2x_1 h_1 h_2 + h_1^2 x_2 + h_1^2 h_2 + x_2 + h_2 \\ &= \underbrace{x_1^2 x_2 + x_2}_{f(x)} + \underbrace{2x_1 x_2 h_1 + x_1^2 h_2 + h_2}_{\operatorname{grad} f(x)\, h} + \underbrace{2x_1 h_1 h_2 + x_2 h_1^2 + h_1^2 h_2}_{\varphi(h)}. \end{aligned}$$

```
f(x,y) = x^2*y+y
x = -4:0.1:4
y = -4:0.1:4
heatmap(x, y, f)
```



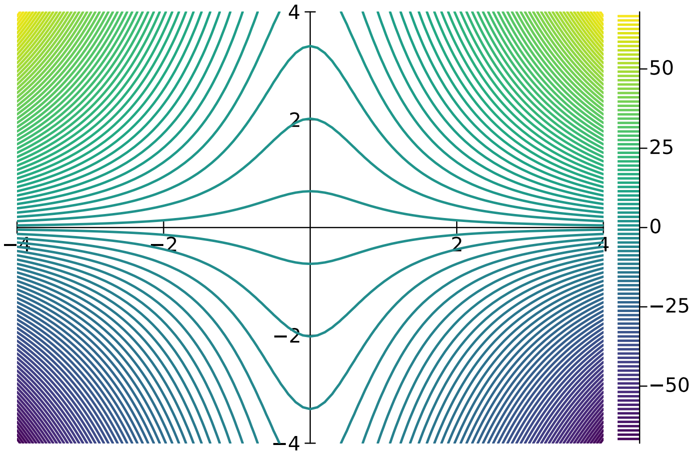
The gradient assigns a vector to each point (x, y).

```
gradf(x,y) = [2*x*y  x^2+1]

xq = range(-4, 4, length=13)
yq = range(-4, 4, length=13)

U = [gradf(x,y)[1] for y in yq, x in xq] *.1
V = [gradf(x,y)[2] for y in yq, x in xq] *.1

X = [x for y in yq, x in xq]
Y = [y for y in yq, x in xq]

quiver!(X, Y; quiver=(U, V), color=:black, linewidth=1.2)
```



We can also visualize the level sets of f.

```
contour(x, y, f,levels = 100)
```



Switching from the vector space setting to coordinates facilitates the actual computation of the derivative. The derivative of a differentiable function is the gradient.

Theorem 11.1.11 (Derivative and gradient in $\mathbb{R}^n$)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at $x \in \mathbb{R}^n$. Then f is partially differentiable at x and*

$$f'(x) = \operatorname{grad} f(x).$$

In particular, $f'(x)e_j = \partial_j f(x)$, for $j = 1, \dots, n$.

Proof. Since f is differentiable at x,

$$f(x+h) = f(x) + f'(x)h + o(\|h\|) \qquad (h \to 0).$$

Fix $j \in \{1, \dots, n\}$ and set $h = te_j$. Then, as $t \to 0$,

$$\frac{f(x+te_j) - f(x)}{t} = \frac{f'(x)(te_j)}{t} + \frac{o(\|te_j\|)}{t} = f'(x)e_j + \frac{o(|t|)}{t}.$$

Hence the limit exists and equals $f'(x)e_j$, i.e.

$$\operatorname{grad} f(x)e_j = \partial_j f(x) = \lim_{t\to 0} \frac{f(x+te_j) - f(x)}{t} = f'(x)e_j.$$

Thus, the two linear maps $\operatorname{grad} f(x)$ and $f'(x)$ coincide, because they are equal on the basis $\{e_1, \dots, e_n\}$. $\square$

The gradient points in the direction of "maximal change".

Example 11.1.12. *For differentiable $f : \mathbb{R}^n \to \mathbb{R}$, the gradient satisfies*

$$f(x+h) = f(x) + \operatorname{grad} f(x)h + o(\|h\|).$$

The Cauchy-Schwarz inequality implies

$$|\operatorname{grad} f(x)\, h| = |\langle \nabla f(x), h\rangle| \le \|\nabla f(x)\|\|h\|$$

with equality if and only if $h = \alpha \nabla f(x)$ for some $\alpha \in \mathbb{R}$. In that case

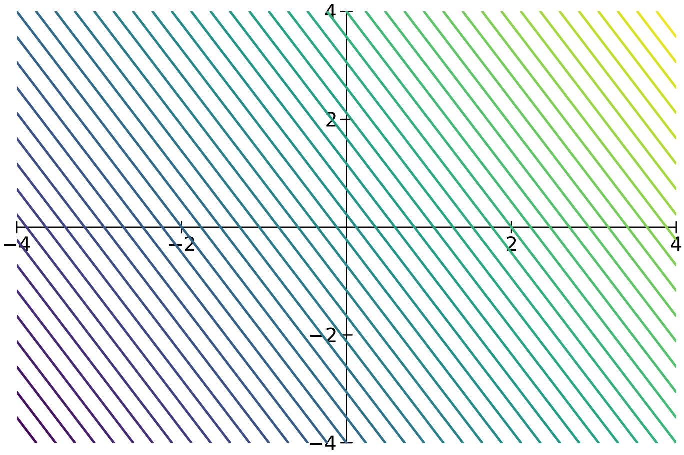
$$\operatorname{grad} f(x)\, h = \alpha \|\nabla f(x)\|^2.$$

For $\alpha > 0$ the vector h points in the direction of the gradient and then f is increasing in this direction since $\operatorname{grad} f(x)\, h > 0$. For $\alpha < 0$, we have $\operatorname{grad} f(x)\, h < 0$ and, hence, f decreases in this direction.

Thus, the local affine linear approximation of $f : \mathbb{R}^n \to \mathbb{R}$ in x_0 is

$$L_f(x) = f(x_0) + \operatorname{grad} f(x_0)(x - x_0).$$

```
(x0,y0) = (1,1)
Lf(x,y) = f(x0,y0)+only(gradf(x0,y0)*[x-x0 ; y-y0])    # only([x]) = x
contour(x, y, Lf,levels = 50, colorbar = false)
```



Alternatively, we may write the local affine linear approximation of f in x_0 as follows:

```
using ForwardDiff
f(x) = x[1]^3 - 3x[1] + x[1]*x[2]^2

function myFirstOrderApproximation(f,x0)
    ∇f = ForwardDiff.gradient(f,x0)
    return x -> f(x0)+∇f'*(x-x0)
end
```

Every differentiable function is partially differentiable. The reverse implication holds under an additional continuity assumption.

Theorem 11.1.13. *Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$. If f is partially differentiable in U and all partial derivatives $\partial_1 f, \ldots, \partial_n f$ are continuous at $x \in U$, then f is differentiable at x and*

$$f'(x) = \operatorname{grad} f(x).$$

Proof. Let $x \in U$ and $h = (h_1, \ldots, h_n) \in \mathbb{R}^n$ with $x + h \in U$. Define points

$$x^{(j)} := x + \sum_{i=1}^{j} h_i e_i, \qquad j = 0, \ldots, n.$$

The telescope sum leads to

$$f(x+h) - f(x) = \sum_{j=1}^{n} \big(f(x^{(j)}) - f(x^{(j-1)})\big).$$

Consider the one-dimensional functions

$$\varphi_j(t) := f(x^{(j-1)} + te_j), \qquad j = 1, \ldots, n,$$

defined for t near 0. The partial differentiability yields

$$\varphi_j'(t) = \lim_{\delta \to 0} \frac{f(x^{(j-1)} + (t+\delta)e_j) - f(x^{(j-1)} + te_j)}{\delta} = \partial_j f(x^{(j-1)} + te_j).$$

By the mean value theorem, there exists $\theta_j \in (0,1)$ such that

$$\begin{aligned} f(x^{(j)}) - f(x^{(j-1)}) = \varphi_j(h_j) - \varphi_j(0) &= \varphi_j'(\theta_j h_j) h_j \\ &= \partial_j f(x^{(j-1)} + \theta_j h_j e_j)\, h_j. \end{aligned}$$

We now obtain

$$f(x+h) - f(x) = \sum_{j=1}^{n} \partial_j f(x^{(j-1)} + \theta_j h_j e_j)\, h_j.$$

This implies

$$f(x+h) = f(x) + \sum_{j=1}^{n} \partial_j f(x)\, h_j + \varphi(h),$$

where the remainder $\varphi(h)$ is

$$\begin{aligned} \varphi(h) &= \sum_{j=1}^{n} \partial_j f(x^{(j-1)} + \theta_j h_j e_j)\, h_j - \sum_{j=1}^{n} \partial_j f(x)\, h_j \\ &= \sum_{j=1}^{n} \Big(\partial_j f(x^{(j-1)} + \theta_j h_j e_j) - \partial_j f(x)\Big)\ h_j. \end{aligned}$$

To verify $\varphi(h)/\|h\| \to 0$, we apply Cauchy-Schwarz in $\mathbb{R}^n$,

$$\begin{aligned} |\varphi(h)| &\le \sum_{j=1}^{n} \Big|\Big(\partial_j f(x^{(j-1)} + \theta_j h_j e_j) - \partial_j f(x)\Big)\ h_j\Big| \\ &\le \|h\| \Big(\sum_{j=1}^{n} \Big|\Big(\partial_j f(x^{(j-1)} + \theta_j h_j e_j) - \partial_j f(x)\Big)\Big|^2\Big)^{1/2}. \end{aligned}$$

For $h \to 0$, we note that $x^{(j-1)} + \theta_j h_j e_j \to x$, so that continuity of $\partial_j f$ implies

$$\partial_j f(x^{(j-1)} + \theta_j h_j e_j) \to \partial_j f(x).$$

This leads to $\varphi(h)/\|h\| \to 0$ and hence f is differentiable in x with

$$f(x+h) = f(x) + \sum_{j=1}^{n} \partial_j f(x)\, h_j + \varphi(h) = f(x) + \operatorname{grad} f(x) h + \varphi(h).$$

□

Every function $f : \mathbb{R}^n \to \mathbb{R}^m$ can be written componentwise as

$$f(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_m(x) \end{pmatrix}, \qquad f_i : \mathbb{R}^n \to \mathbb{R}\,.$$

Remark 11.1.14 (Jacobian matrix)**.** *Since convergence in $\mathbb{R}^m$ means convergence in each component, the function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if and only if all f_i, $i = 1, \ldots, m$ are differentiable. In that case, the Jacobian matrix*

$$J_f(x) := \begin{pmatrix} \operatorname{grad} f_1(x) \\ \vdots \\ \operatorname{grad} f_m(x) \end{pmatrix} = \big(\partial_j f_i(x)\big)_{i,j} \in \mathbb{R}^{m\times n}\,.$$

is the derivative $f'(x) = J_f(x)$.

If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable on an open set U, then the mapping

$$x \mapsto f'(x) = J_f(x) \in \mathbb{R}^{m\times n}$$

is continuous if and only if all partial derivatives $x \mapsto \partial_j f_i(x)$ are continuous. In this case, f is called *continuously differentiable.*

We say that $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable if $\operatorname{grad} f : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. In other words, if the map $\partial_j f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable for all $j = 1, \ldots, n$.

Lemma 11.1.15 (mixed partial derivatives commute)**.** *Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$ be twice continuously partially differentiable. Then for all $i, j \in \{1, \ldots, n\}$ and all $x \in U$,*

$$\partial_i \partial_j f(x) = \partial_j \partial_i f(x).$$

Proof. Fix $x \in U$ and indices i, j. Since U is open, there exists $\varepsilon > 0$ such that

$$x + se_i + te_j \in U \qquad \text{for all } |s|, |t| < \varepsilon.$$

Define the function $\varphi : \mathbb{R}^2 \to \mathbb{R}$ by

$$\varphi(s,t) := f(x + se_i + te_j).$$

Since f is twice continuously partially differentiable, the function φ has continuous second-order partial derivatives, and by the chain rule we have

$$\partial_1\partial_2\varphi(s,t) = \partial_i\partial_j f(x + se_i + te_j), \qquad \partial_2\partial_1\varphi(s,t) = \partial_j\partial_i f(x + se_i + te_j).$$

It therefore suffices to show that

$$\partial_1\partial_2\varphi(0,0) = \partial_2\partial_1\varphi(0,0).$$

Using the one-dimensional mean value theorem twice, there are $\xi_s \in (0,s)$ and $\xi_t \in (0,t)$, such that

$$\underbrace{\varphi(s,t)-\varphi(0,t)}_{p(t)} - \big(\underbrace{\varphi(s,0)-\varphi(0,0)}_{p(0)}\big) = p'(\xi_t)t = \partial_2\left(\varphi(s,\xi_t)-\varphi(0,\xi_t)\right)t$$
$$= \left(\partial_2\varphi(s,\xi_t)-\partial_2\varphi(0,\xi_t)\right)t$$
$$= \partial_1\partial_2\varphi(\xi_s,\xi_t)st.$$

We repeat this process with interchanged roles of s and t. There are $\eta_s \in (0,s)$ and $\eta_t \in (0,t)$, such that

$$\underbrace{\varphi(s,t)-\varphi(s,0)}_{q(s)} - \big(\underbrace{\varphi(0,t)-\varphi(0,0)}_{q(0)}\big) = q'(\eta_s)s = \partial_1\left(\varphi(\eta_s,t)-\varphi(\eta_s,0)\right)s$$
$$= \left(\partial_1\varphi(\eta_s,t)-\partial_1\varphi(\eta_s,0)\right)s$$
$$= \partial_2\partial_1\varphi(\eta_s,\eta_t)st.$$

We have shown that

$$\partial_1\partial_2\varphi(\xi_s,\xi_t) = \partial_2\partial_1\varphi(\eta_s,\eta_t).$$

The limit $(s,t)\to 0$ implies $(\xi_s,\xi_t),(\eta_s,\eta_t)\to 0$. The continuity of $\partial_i\partial_j f$ and $\partial_j\partial_i f$ yields continuity of $\partial_1\partial_2\varphi$ and $\partial_2\partial_1\varphi$ in 0, so that we arrive at

$$\partial_1\partial_2\varphi(0,0) = \partial_2\partial_1\varphi(0,0).$$

□

For $f:\mathbb{R}^n\to\mathbb{R}$, the gradient $\operatorname{grad} f$ collects first-order (partial) derivatives. Second-order differential information is collected by the Hessian.

Definition 11.1.16 (Hessian matrix)**.** *Let $U\subseteq\mathbb{R}^n$ be open and suppose that $f:U\to\mathbb{R}$ is twice continuously differentiable. The* Hessian matrix *of f at $x\in U$ is defined by*

$$H_f(x) := \left(\partial_i\partial_j f(x)\right)_{i,j=1}^n.$$

Since f is twice continuously differentiable, the mixed partial derivatives commute, and hence the Hessian matrix is symmetric. When it comes to local extrema in the next section, it plays the role of the second derivative of univariate functions to decide if a local extremum is a local maximum or minimum.

Example 11.1.17 (Hessian)**.** *Consider the function $f:\mathbb{R}^2\to\mathbb{R}$ defined by*

$$f(x) = x_1^3 + x_2^2\exp(x_1).$$

We compute

$$\begin{aligned} \partial_1 f(x) &= 3x_1^2 + x_2^2\exp(x_1), & \partial_2 f(x) &= 2x_2\exp(x_1)\\ \partial_1\partial_1 f(x) &= 6x_1 + x_2^2\exp(x_1), & \partial_2\partial_2 f(x) &= 2\exp(x_1)\\ \partial_2\partial_1 f(x) &= 2x_2\exp(x_1), & \partial_1\partial_2 f(x) &= 2x_2\exp(x_1). \end{aligned}$$

The Hessian matrix is

$$H_f(x) = \begin{pmatrix} 6x_1 + x_2^2 \exp(x_1) & 2x_2 \exp(x_1) \\ 2x_2 \exp(x_1) & 2\exp(x_1) \end{pmatrix}.$$

Essentials

Pairs of dual frames extend the concept of orthonormal bases and provide structured coordinate representations. Through these representations, continuous maps on Hilbert spaces can be discretized. Truncating the discretized representation yields finite-dimensional approximations of the original map. If f is defined on finite-dimensional coordinate-free normed spaces, then, after choosing bases, the Fréchet derivative of f corresponds to the Fréchet derivative of its coordinate representation. Fréchet differentiability in coordinates reduces to the existence of partial derivatives. You thus learned that differentiability in $\mathbb{R}^n$ is expressed in terms of partial derivatives, gradients, and Jacobian matrices. Moreover, mixed partial derivatives commute under suitable regularity assumptions.

RTA Continuous maps are represented in coordinates via dual frames. Truncation provides finite-dimensional approximations. Representation by coordinates turns differentiability into partial derivatives. Gradients and Jacobians lead to local linear approximations.

Hilbert space, Riesz representation theorem, adjoint operator, orthonormal basis, pairs of dual frames, partial derivatives, gradient, Jacobian matrix, Hessian matrix

11.2 Taylor's formula and local extrema

This section develops Taylor's formula for multivariate functions as a local polynomial approximation and uses it to study local extrema. We derive practical criteria for local extrema based on gradients and Hessians.

First we introduce higher-order partial derivatives and derive Taylor's formula in coordinate spaces. This will allow us to classify local extrema using second-order information.

For $k \in \mathbb{N}$ and $j \in \{1, \ldots, n\}$, we define the k-th order partial derivative by

$$\partial_j^k := \underbrace{\partial_j \cdots \partial_j}_{k \text{ times}}.$$

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, we define

$$\partial^\alpha := \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}, \qquad |\alpha| := \alpha_1 + \cdots + \alpha_n, \qquad \alpha! := \alpha_1! \cdots \alpha_n!.$$

For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $y = (y_1, \ldots, y_n) \in \mathbb{R}^n$, we write

$$(x - y)^\alpha := (x_1 - y_1)^{\alpha_1} \cdots (x_n - y_n)^{\alpha_n}.$$

Theorem 11.2.1 (Taylor formula in $\mathbb{R}^n$). *Let $U \subseteq \mathbb{R}^n$ be open, $k \in \mathbb{N}$, and let $f : U \to \mathbb{R}$*

be k times continuously differentiable. Then for every $x_0 \in U$,

$$f(x) = \sum_{|\alpha| \le k} \frac{\partial^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha + o(\|x - x_0\|^k) \qquad (x \to x_0).$$

The Taylor polynomial,

$$x \mapsto \sum_{|\alpha| \le k} \frac{\partial^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha,$$

provides a polynomial approximation of f near x_0.

Remark 11.2.2 (Local approximation). *In data analysis, Taylor expansions are used to approximate complex models locally by simpler ones.*

Proof. Set $h := x - x_0$ and assume $\|h\|$ is small enough so that the line segment $\{x_0 + th : t \in [0,1]\} \subseteq U$. Define the one-dimensional function

$$g : [0,1] \to \mathbb{R}, \qquad g(t) := f(x_0 + th).$$

Step 1: compute $g^{(j)}(0)$ in multi-index form. For $j \in \{0, \dots, k\}$ one proves by induction (using the chain rule and the commutation of mixed partial derivatives) that g is k-times continuously differentiable and

$$g^{(j)}(t) = \sum_{|\alpha| = j} \frac{j!}{\alpha!} (\partial^\alpha f)(x_0 + th)\, h^\alpha, \qquad t \in [0,1]. \tag{11.1}$$

In particular,

$$\frac{g^{(j)}(0)}{j!} = \sum_{|\alpha| = j} \frac{(\partial^\alpha f)(x_0)}{\alpha!} h^\alpha.$$

Summing over $j = 0, \dots, k$ yields

$$\sum_{j=0}^{k} \frac{g^{(j)}(0)}{j!} = \sum_{|\alpha| \le k} \frac{(\partial^\alpha f)(x_0)}{\alpha!} h^\alpha,$$

which is exactly the Taylor polynomial of order k.

Step 2: The univariate Taylor formula with integral remainder. The univariate Taylor formula for g yields

$$g(1) = \sum_{j=0}^{k-1} \frac{g^{(j)}(0)}{j!} + \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} g^{(k)}(t)\, dt.$$

We add and subtract $g^{(k)}(0)$ inside the integral and use $\int_0^1 (1-t)^{k-1}\, dt = 1/k$ to obtain

$$g(1) = \sum_{j=0}^{k} \frac{g^{(j)}(0)}{j!} + \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} \big(g^{(k)}(t) - g^{(k)}(0)\big)\, dt.$$

Since $g(1) = f(x)$, the remainder

$$R(h) := \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} \big(g^{(k)}(t) - g^{(k)}(0)\big)\, dt.$$

leads to the formula

$$f(x) = \sum_{j=0}^{k} \frac{g^{(j)}(0)}{j!} + R(h).$$

Step 3: estimate the remainder. We use the representation (11.1) for $j = k$ and derive

$$g^{(k)}(t) - g^{(k)}(0) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} \Big((\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big) h^\alpha.$$

Hence, with $|h^\alpha| \le \|h\|^{|\alpha|} = \|h\|^k$,

$$|g^{(k)}(t) - g^{(k)}(0)| \le \|h\|^k \max_{|\alpha|=k} \Big| (\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big| \sum_{|\alpha|=k} \frac{k!}{\alpha!}.$$

Thus, there is a constant $C > 0$ that depends only on k and n such that

$$|R(h)| \le C\,\|h\|^k \sup_{t\in[0,1]} \max_{|\alpha|=k} \Big| (\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big|.$$

Since each $\partial^\alpha f$ with $|\alpha| = k$ is continuous at x_0, the function

$$\omega(h) := \sup_{t\in[0,1]} \max_{|\alpha|=k} \Big| (\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big| \to 0$$

as $h \to 0$. Hence $R(h) = o(\|h\|^k)$. □

Example 11.2.3 (second-order Taylor expansion)**.** *For $k = 2$, Taylor's formula can be written in the compact form*

$$f(x) = f(x_0) + \operatorname{grad} f(x_0)(x - x_0) + \frac{1}{2}(x - x_0)^\top H_f(x_0)(x - x_0) + o(\|x - x_0\|^2).$$

```julia
using ForwardDiff
f(x) = x[1]^2 +x[1]*x[2]+2*x[2]^2

function mySecondOrderApproximation(f,x0)
    ∇f = ForwardDiff.gradient(f,x0)
    Hf = ForwardDiff.hessian(f, x0)
    return x -> f(x0)+∇f'*(x-x0)+1/2*(x-x0)'*Hf*(x-x0)
end
```

Example 11.2.4. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x) = x_1^2 + x_1 x_2 + 2x_2^2.$$

At the point $x_0 = (0,0)$ we compute $f(0,0) = 0$ and

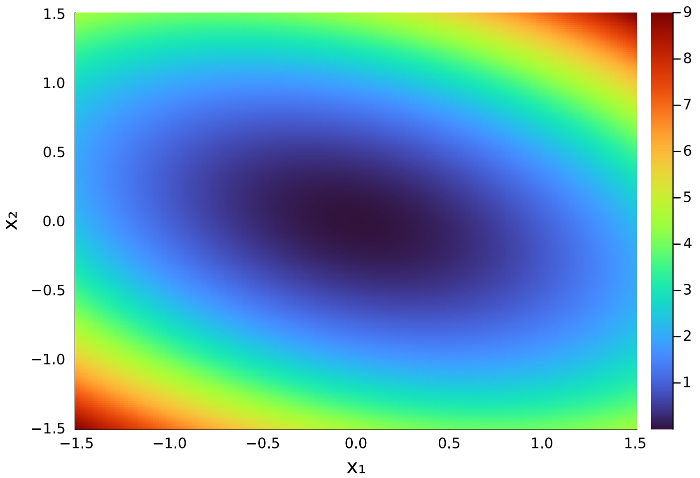
$$\operatorname{grad} f(0,0) = (0,0), \qquad H_f(0,0) = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

Thus the second-order Taylor polynomial of f at $(0,0)$ is given by

$$
\begin{aligned}
f(x) = \frac{1}{2}(x_1, x_2)\begin{pmatrix}2 & 1\\ 1 & 4\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} &= \frac{1}{2}(x_1, x_2)\begin{pmatrix}2x_1 + x2\\ x_1 + 4x_2\end{pmatrix}\\
&= \frac{1}{2}\left(2x_1^2 + x_1x_2 + x_1x_2 + 4x_2^2\right)\\
&= x_1^2 + x_1x_2 + 2x_2^2.
\end{aligned}
$$

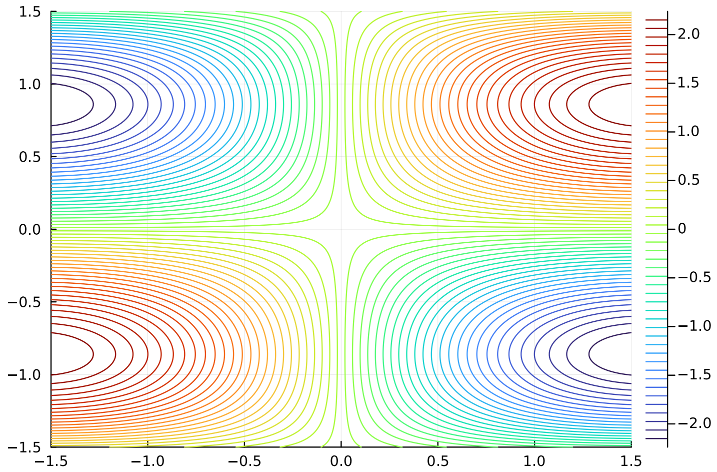
It coincides with f, so that the polynomial approximation is exact in this case.

```
x1 = range(-1.5, 1.5, length=200)
x2 = range(-1.5, 1.5, length=200)

F2(x) = mySecondOrderApproximation(f,[0;0])(x)

Z = [F2([x;y]) for y in x2, x in x1]
heatmap(x1, x2, Z, xlabel="$x_1$", ylabel="$x_2$")
```

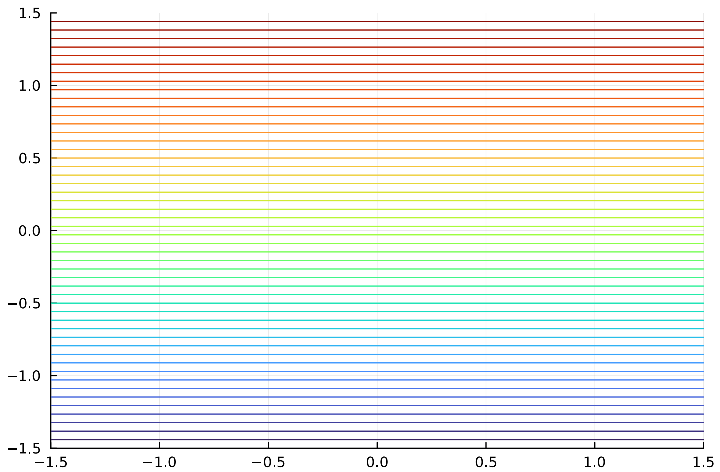


We also consider $f(x) = 4\sin(x_1)\cos(x_2)x_2$ and compute the first and second order approximation in $x_0 = (1,0)^\top$.

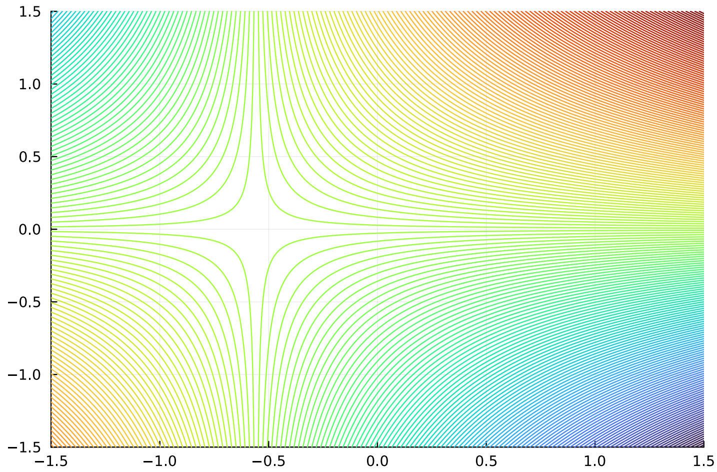
```
f(x) = 4*sin(x[1])*cos(x[2])*x[2]
x1 = range(-1.5, 1.5, length=200)
x2 = range(-1.5, 1.5, length=200)

Z = [f([x;y]) for y in x2, x in x1]

contour(x1, x2, Z,levels = 50)
```




```
x0 = [1;0]

F1(x) = myFirstOrderApproximation(f,x0)(x)

Z = [F1([x;y]) for y in x2, x in x1]
contour(x1, x2, Z,levels = 50,colorbar=false)
```




```
F2(x) = mySecondOrderApproximation(f,x0)(x)

Z = [F2([x;y]) for y in x2, x in x1]

contour(x1, x2, Z,levels = 200, colorbar=false)
```



Recall that a matrix $A \in \mathbb{R}^{n\times n}$ is called positive definite if

$$x^\top Ax > 0, \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}.$$

It is called negative definite if $-A$ is positive definite and indefinite if there are $x, y \in \mathbb{R}^n$ such that

$$x^\top Ax < 0, \qquad y^\top Ay > 0.$$

The gradient and the Hessian combined provide a sufficient condition for local extrema.

Theorem 11.2.5 (Second-order sufficient conditions for extrema)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable and let $x_0 \in \mathbb{R}^n$ be a critical point, i.e.* $\operatorname{grad} f(x_0) = 0$.

(i) If $H_f(x_0)$ is positive definite, then x_0 is a strict local minimum.

(ii) If $H_f(x_0)$ is negative definite, then x_0 is a strict local maximum.

(iii) If $H_f(x_0)$ is indefinite, then x_0 is a saddle point.

Remark 11.2.6. *If the Hessian $H_f(x_0)$ is positive semidefinite or negative semidefinite but not definite, the second-order test is inconclusive.*

Remark 11.2.7 (Local extrema)**.** *In data-driven models, local extrema correspond to locally optimal parameter values where small changes do not improve the objective function.*

Proof. Since f is twice continuously differentiable, Taylor's formula of second order yields, for h sufficiently small,

$$f(x_0 + h) = f(x_0) + \operatorname{grad} f(x_0) \cdot h + \frac{1}{2} h^\top H_f(x_0) h + o(\|h\|^2).$$

Because x_0 is a critical point, $\operatorname{grad} f(x_0) = 0$, and hence

$$f(x_0 + h) = f(x_0) + \frac{1}{2} h^\top H_f(x_0) h + \varphi(h),$$

where $\varphi(h)/\|h\|^2 \to 0$ for $h \to 0$.

(i) The idea is to verify that $\frac{1}{2}h^\top H_f(x_0)h$ is positive and larger than $\varphi(h)$. We first define

$$\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}.$$

As the pre-image of the closed set $\{1\}$ of a continuous map $\|\cdot\|$, the set $\mathbb{S}^{n-1}$ is closed. It is also bounded. Heine-Borel extends to from $\mathbb{R}$ to $\mathbb{R}^n$ and therefore $\mathbb{S}^{n-1}$ is compact. Since $h \mapsto \frac{1}{2}h^\top H_f(x_0)h$ is continuous, its image of $\mathbb{S}^{n-1}$ is compact. Therefore, it assumes its minimum, which must be a positive number since $H_f(x_0)$ is positive definite. We define

$$\alpha := \min_{h\in\mathbb{S}^{n-1}} \frac{1}{2}h^\top H_f(x_0)h.$$

For $h \in \mathbb{R}^n \setminus \{0\}$, we calculate

$$\frac{1}{2}h^\top H_f(x_0)h = \frac{1}{2}\|h\|\frac{h^\top}{\|h\|}H_f(x_0)\frac{h}{\|h\|}\|h\| \geq \frac{1}{2}\alpha\|h\|^2.$$

Since $\varphi(h)/\|h\|^2 \to 0$ for $h \to 0$, there is $\delta > 0$ such that

$$\frac{|\varphi(h)|}{\|h\|^2} \leq \frac{\alpha}{4}, \qquad \forall \|h\| \leq \delta.$$

For $\|h\| \leq \delta$, we obtain

$$\begin{aligned} f(x_0+h) &= f(x_0) + \frac{1}{2}h^\top H_f(x_0)h + \varphi(h) \\ &\geq f(x_0) + \frac{\alpha}{2}\|h\|^2 - \frac{\alpha}{4}\|h\|^2 \\ &= f(x_0) + \frac{\alpha}{4}\|h\|^2. \end{aligned}$$

This shows that x_0 is a strict local minimum.
(ii) If $H_f(x_0)$ is negative definite, the same argument applied to $-f$ shows that x_0 is a strict local maximum.
(iii) Finally, if $H_f(x_0)$ is indefinite, there exist vectors $u, v \in \mathbb{R}^n$ such that

$$u^\top H_f(x_0)u > 0 \quad \text{and} \quad v^\top H_f(x_0)v < 0.$$

Choosing $x = x_0 + tu$ and $x = x_0 + tv$ with t sufficiently small and using the Taylor expansion shows that $f(x) - f(x_0)$ takes both positive and negative values arbitrarily close to x_0. Hence x_0 is a saddle point. □

Example 11.2.8. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x_1, x_2) = x_1^2 + x_2^2.$$

Then $\mathrm{grad}(f)(x) = (2x_1, 2x_2)$, *hence the only critical point is* $x_0 = (0, 0)$. *Moreover,*

$$H_f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

which is positive definite. Thus $(0, 0)$ *is a strict local minimum.*

Let us revisit Example 10.4.7.

Example 11.2.9. *The function* $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$ *has the gradient and Hessian*

$$\operatorname{grad} f(x) = (x_2\,,\, x_1)\,, \qquad H_f(x) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The gradient vanishes for $x = 0$, *and the Hessian is indefinite due to*

$$(x_1, x_2)\, H_f(x) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2x_1x_2.$$

Thus, $x = 0$ *is a saddle point as already stated in Example 10.4.7.*

Computing extrema by gradient descent

Suppose we want to minimize a convex function $f : \mathbb{R}^n \to \mathbb{R}$. We obtain its minimum by a fixed point iteration.

Convexity plays a key role, and we first characterize convexity by the Hessian matrix. Recall that a matrix $A \in \mathbb{R}^{n\times n}$ is called positive semi-definite if

$$x^\top Ax \geq 0, \qquad \forall\, x \in \mathbb{R}^n\,.$$

Lemma 11.2.1 (Second-order characterization of convexity)**.** *Let* $f \in \mathcal{C}^2(\mathbb{R}^n)$. *Then* f *is convex if and only if* $H_f(x)$ *is positive semi-definite for all* $x \in \mathbb{R}^n$.

The first order characterization of convexity in Lemma 10.4.2 yields

$$f(x) - f(y) \leq f'(x)\,(x-y), \qquad x, y \in \mathbb{R}^n\,.$$

Moreover, we rewrite convexity as follows.

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2), \qquad x_1, x_2 \in \mathbb{R}^n, \quad \lambda_1, \lambda_2 \in [0,1], \quad \lambda_1 + \lambda_2 = 1.$$

The extension to convex linear combinations of length m is known as Jensen's inequality.

Lemma 11.2.2 (Jensen's inequality)**.** *Let* $f : \mathbb{R}^n \to \mathbb{R}$ *be convex and* $m \in \mathbb{N}$ *with* $m \geq 1$. *Then*

$$f(\sum_{i=1}^m \lambda_i x_i) \leq \sum_{i=1}^m \lambda_i f(x_i), \qquad x_i \in \mathbb{R}^n, \quad \lambda_i \in [0,1], \quad \sum_{i=1}^m \lambda_i = 1.$$

Proof. The case $m = 1$ is obvious and $m = 2$ is the definition of convexity. Assume that the claim holds for m. To verify the inequality for $m+1$, we consider $\lambda_i \in [0,1]$ and $\sum_{i=1}^{m+1} \lambda_i = 1$. There must be one index i such $\lambda_i \neq 0$. Without loss of generality, we may assume that $i = m+1$, otherwise we may reorder. For $\lambda_{m+1} \neq 0$, we obtain

$$
\begin{aligned}
f(\sum_{i=1}^{m+1} \lambda_i x_i) &= f((1-\lambda_{m+1}) \underbrace{\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} x_i}_{\tilde{x}_1} + \lambda_{m+1} x_{m+1}) \\
&\leq (1-\lambda_{m+1}) f(\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} x_i) + \lambda_{m+1} f(x_{m+1}).
\end{aligned}
$$

The induction hypothesis for m leads to

$$
f(\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} x_i) \leq \sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} f(x_i),
$$

because $\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} = \frac{1}{1-\lambda_{m+1}} \sum_{i=1}^{m} \lambda_i = 1$. We therefore derive

$$
\begin{aligned}
f(\sum_{i=1}^{m+1} \lambda_i x_i) &\leq (1-\lambda_{m+1}) \sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} f(x_i) + \lambda_{m+1} f(x_{m+1}) \\
&= \sum_{i=1}^{m+1} \lambda_i f(x_i).
\end{aligned}
$$

□

Gradient descent

Theorem 11.2.3 (Gradient descent). *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable and assume that*

$$
\|\nabla f(x)\| \leq C \qquad \text{for all } x \in \mathbb{R}^d
$$

for some $C > 0$. Let x^* be a minimizer of f. For an initial $x_0 \in \mathbb{R}^d$, consider gradient descent

$$
x_{k+1} := x_k - \eta \nabla f(x_k)
$$

with step size $\eta > 0$. Define the average iterate

$$
\bar{x}_N := \frac{1}{N} \sum_{k=0}^{N-1} x_k.
$$

Then for every $N \geq 1$,

$$
|f(\bar{x}_N) - f(x^*)| \leq \frac{\|x_0 - x^*\|^2}{2\eta N} + \frac{\eta C^2}{2}.
$$

In particular, with $\eta = \|x_0 - x^*\|/(C\sqrt{N})$,

$$
|f(\bar{x}_N) - f(x^*)| \leq N^{-\frac{1}{2}} C \|x_0 - x^*\|.
$$

Thus, the error is bounded by $|f(\bar{x}_N) - f(x^*)|$ decays as $N^{-1/2}$.

Proof. Let $e_k := x_k - x^*$. Expanding the update gives

$$\begin{aligned} \|e_{k+1}\|^2 &= \|x_k - \eta\nabla f(x_k) - x^*\|^2 \\ &= \|e_k - \eta\nabla f(x_k)\|^2 \\ &= \|e_k\|^2 - 2\eta\langle\nabla f(x_k), e_k\rangle + \eta^2\|\nabla f(x_k)\|^2, \end{aligned}$$

and rearranging terms leads to

$$2\eta\langle\nabla f(x_k), e_k\rangle = \|e_k\|^2 - \|e_{k+1}\|^2 + \eta^2\|\nabla f(x_k)\|^2.$$

The first-order characterization of convexity in Lemma 10.4.2 yields

$$f(x_k) - f(x^*) \leq \operatorname{grad} f(x_k)\,(x_k - x^*) = \langle\nabla f(x_k), e_k\rangle.$$

The gradient bound, $\|\nabla f(x_k)\|^2 \leq C^2$, leads to

$$2\eta\big(f(x_k) - f(x^*)\big) \leq \|e_k\|^2 - \|e_{k+1}\|^2 + \eta^2 C^2.$$

Summing from $k = 0$ to $N - 1$ telescopes:

$$\begin{aligned} 2\eta\sum_{k=0}^{N-1}\big(f(x_k) - f(x^*)\big) &\leq \sum_{k=0}^{N-1}(\|e_k\|^2 - \|e_{k+1}\|^2) + N\eta^2C^2 \\ &= \|e_0\|^2 - \|e_{N+1}\|^2 + N\eta^2C^2 \\ &\leq \|e_0\|^2 + N\eta^2C^2. \end{aligned}$$

Divide by $2\eta N$ and use convexity (Jensen) $f(\bar{x}_N) \leq \frac{1}{N}\sum_{k=0}^{N-1} f(x_k)$ to conclude the claim. $\square$

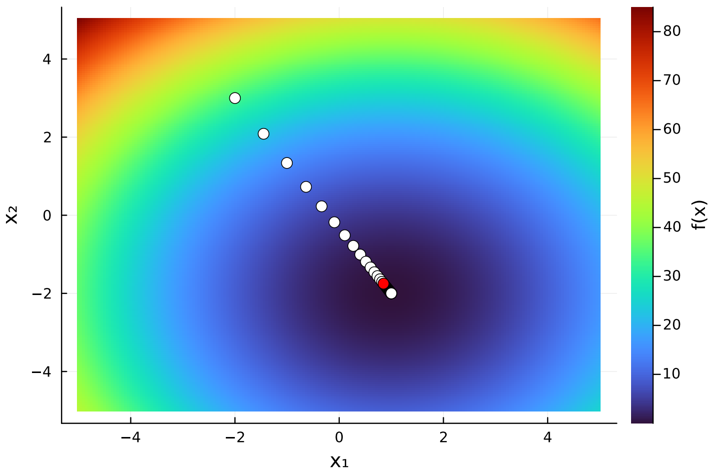
To implement gradient descent, we store all iterates and return both, the iterates and the average $\bar{x}_N$.

```
function myGradientDescent(f, x0; η = 0.01, N = 1000)
    d = length(x0)
    x_vec = zeros(d,N)
    x_vec[:,1] = x0

    for k in 1:N-1
        x_vec[:,k+1] = x_vec[:,k] - η * ForwardDiff.gradient(f,x_vec[:,k])
    end

    x_avg = sum(x_vec[:,k] for k=1:N)/N

    return x_vec , x_avg
end
```

The function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x_1, x_2) = (x_1 - 1)^2 + (x_2 + 2)^2$ has the minimum in $x^* = (1, -2)^\top$.

```
f(x::Vector) = (x[1] - 1)^2 +(x[2] + 2)^2

x1 = range(-5, 5, length=200)
x2 = range(-5, 5, length=200)
Z = [f([x, y]) for y in x2, x in x1]
heatmap(x1, x2, Z, xlabel="x₁", ylabel="x₂", colorbar_title="f(x)")
```

```
x0 = [-2;3]
x_vec , x_avg = myGradientDescent(f, x0; η = 0.01)

heatmap(x1, x2, Z, xlabel="x₁", ylabel="x₂", colorbar_title="f(x)", c = :turbo)
scatter!(x_vec[1,1:10:end], x_vec[2,1:10:end], color="white", markersize=5)
scatter!(x_avg[1,:], x_avg[2,:], color="red", markersize=5)
```



If f is not convex, then gradient descent converges towards a critical point. To see this, we still need some preparation.

Definition 11.2.4. *A function $g : \mathbb{R}^d \to \mathbb{R}$ is called L-Lipschitz continuous if there exists a constant $L > 0$ such that*

$$\|g(x) - g(y)\| \leq L\|x - y\|, \qquad \forall x, y \in \mathbb{R}^d.$$

Lemma 11.2.5. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and suppose that ∇f is L-Lipschitz continuous. Then*

$$f(y) - f(x) \leq \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2, \qquad \forall x, y \in \mathbb{R}^d.$$

Proof. Define the path $\phi : [0, 1] \to \mathbb{R}$ by

$$\phi(t) = f(x + t(y - x)).$$

The chain rule leads to

$$\phi'(t) = \nabla f(x + t(y - x))^\top (y - x).$$

The fundamental theorem of calculus and Cauchy-Schwarz yield

$$\begin{aligned}
f(y) - f(x) &= \int_0^1 \phi'(t)dt \\
&= \int_0^1 \nabla f(x + t(y-x))^\top (y-x)dt \\
&= \nabla f(x)^\top (y-x) + \int_0^1 \left(\nabla f(x + t(y-x) - \nabla f(x)^\top\right)^\top (y-x)dt \\
&\leq \nabla f(x)^\top (y-x) + \int_0^1 \|\nabla f(x + t(y-x) - \nabla f(x)\| \|y-x\| dt \\
&\leq \nabla f(x)^\top (y-x) + \int_0^1 L\|t(y-x)\| \|y-x\| dt \\
&= \nabla f(x)^\top (y-x) + L\|y-x\|^2 \int_0^1 tdt \\
&= \nabla f(x)^\top (y-x) + \frac{L}{2}\|y-x\|^2.
\end{aligned}$$

□

Theorem 11.2.6. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable, bounded from below, so that $f(x) \geq r_0$ for all $x \in \mathbb{R}^d$. Suppose that ∇f is L-Lipschitz continuous. Then gradient descent*

$$x_{k+1} = x_k - \eta \nabla f(x_k), \qquad 0 < \eta < \frac{2}{L},$$

satisfies

$$\nabla f(x_k) \to 0.$$

Proof. By Lemma 11.2.5,

$$f(x_{k+1}) - f(x_k) \leq \nabla f(x_k)^\top (x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2.$$

According to $x_{k+1} = x_k - \eta \nabla f(x_k)$, we derive

$$f(x_{k+1}) - f(x_k) \leq -\eta \|\nabla f(x_k)\|^2 + \frac{L\eta^2}{2}\|\nabla f(x_k)\|^2.$$

We define $c := \eta(1 - \frac{L\eta}{2}) > 0$ and obtain

$$f(x_{k+1}) - f(x_k) \leq -\eta(1 - \frac{L\eta}{2})\|\nabla f(x_k)\|^2 = -c\|\nabla f(x_k)\|^2.$$

This implies $c\|\nabla f(x_k)\|^2 \leq f(x_k) - f(x_{k+1})$ and summing from $k = 0$ to N yields

$$c\sum_{k=0}^{N} \|\nabla f(x_k)\|^2 \leq \sum_{k=0}^{N} (f(x_k) - f(x_{k+1})) = f(x_0) - f(x_{N+1}) \leq f(x_0) - r_0.$$

Since this inequality holds for all $N \in \mathbb{N}$, we deduce that the series converges with

$$c\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 \leq f(x_0) - r_0.$$

This implies that $\|\nabla f(x_k)\|^2 \to 0$. □

Stochastic gradient descent

If $F : \mathbb{R}^d \to \mathbb{R}$ is an average over simpler components,

$$F(x) := \frac{1}{n}\sum_{i=1}^{n} f_i(x),$$

then we can use alternatives to the gradient of F. Instead of computing

$$\operatorname{grad} F(x) = \frac{1}{n}\sum_{i=1}^{n} \operatorname{grad} f_i(x)$$

in each iteration, we restrict us to only one index $i \in \{1, \dots, n\}$. That is we replace $\operatorname{grad} F(x)$ by $\operatorname{grad} f_i(x)$, where $i \in \{1, \dots, n\}$ is random in each iteration.

Remark 11.2.7. *This makes gradient descent significantly more efficient and turns out to be a game changer in large scale machine learning scenarios.*

Theorem 11.2.8 (Stochastic gradient descent for finite-sum problems)**.** *Let $f_1, \dots, f_n : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable, and define*

$$F(x) := \frac{1}{n}\sum_{i=1}^{n} f_i(x).$$

Let x^* be a minimizer of f. Assume that

$$\|\nabla f_i(x)\| \le C \quad \textit{for all } x \in \mathbb{R}^d,\ i = 1, \dots, n,$$

for some $C > 0$.
Let $(i_k)_{k\in\mathbb{N}}$ be independent uniformly distributed on $\{1, \dots, n\}$, and define

$$x_{k+1} = x_k - \eta\,\nabla f_{i_k}(x_k),$$

with step size $\eta > 0$. Then for $\bar{x}_N := \frac{1}{N}\sum_{k=0}^{N-1} x_k$,

$$\mathbb{E}[F(\bar{x}_N)] - F(x^*) \le \frac{\|x_0 - x^*\|^2}{2\eta N} + \frac{\eta C^2}{2}.$$

In particular, choosing $\eta = \|x_0 - x^*\|/(C\sqrt{N})$ yields

$$\mathbb{E}[F(\bar{x}_N)] - F(x^*) \le N^{-\frac{1}{2}}\, C\|x_0 - x^*\|.$$

Proof idea. The only probabilistic input we use is the following elementary fact: if i is independently, uniformly distributed on $\{1, \dots, n\}$ and $\psi : \{1, \dots, n\} \to \mathbb{R}^m$ is any function, then

$$\mathbb{E}_i[\psi(i)] = \frac{1}{n}\sum_{j=1}^{n} \psi(j).$$

In our setting, this implies

$$\mathbb{E}_{i_k}\left[\nabla f_{i_k}(x)\right] = \frac{1}{N}\sum_{i=1}^{N} \nabla f_i(x) = \nabla F(x). \tag{11.2}$$

We now follow the proof of gradient descent and use the expectation at the appropriate locations. We omit the details. □

```
function myStochasticGradientDescent(F, x0; η = 0.01, N = 10_000)
    d = length(x0)
    n = length(F)
    x_vec = zeros(d,N)
    x_vec[:,1] = x0

    for k in 1:N-1
        i = rand(1:n)
        x_vec[:,k+1] = x_vec[:,k] - η * ForwardDiff.gradient(F[i],x_vec[:,k])
    end

    x_avg = sum(x_vec[:,k] for k=1:N)/N

    return x_vec , x_avg
end

F = [ x -> (x[1] - 1)^2 + (x[2] - 2)^2,
      x -> 2(x[1] - 1)^2 + (x[2] - 2)^2,
      x -> (x[1] - 1)^2 + 3(x[2] - 2)^2 ]

x0 = [10.0; -5.0]

x_vec , x_avg = myStochasticGradientDescent(F,x0;η=0.01, N=100_000)
x_avg
```

```
2-element Vector{Float64}:
 1.003158302447018
 1.997624054388961
```

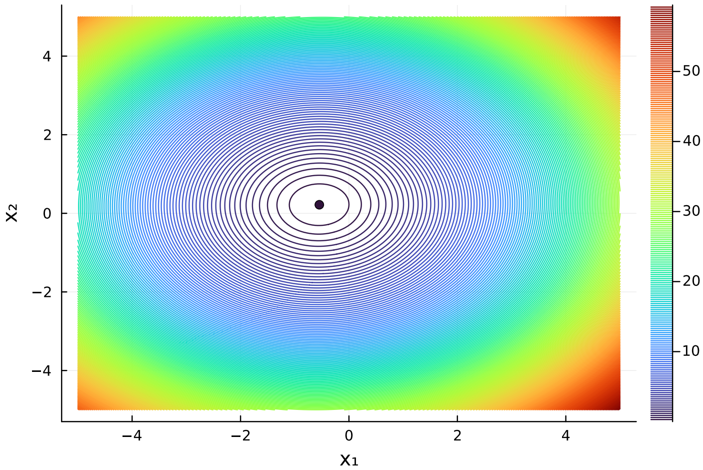
We now provide an example for large n.

```
using Random, ForwardDiff, Statistics
n = 1_000
d = 2

# model setup
Random.seed!(42)
A = randn(n, d)
x_true = randn(d)
y = A * x_true .+ 0.1*randn(n)

F = [x -> (dot(A[i, :],x) - y[i])^2 for i in 1:n]
```

```
x0 = zeros(d)
x_vec, x_avg = myStochasticGradientDescent(F, x0; η=0.01, N=10_000)

println("x_true = ", x_true)
println("x_avg  = ", x_avg)
```

```
x_true = [-0.5445681210073285, 0.2208132624522603]
x_avg = [-0.5467396476613867, 0.21803998455760248]
```

```
g(x) = sum(F[i](x) for i=1:n)/n
Z = [g([x, y]) for y in x2, x in x1]
contour(x1, x2, Z, xlabel="x₁", ylabel="x₂", levels = 200)
scatter!(x_avg[1,:],x_avg[2,:])
```



Mini-batch SGD

Mini-batch stochastic gradient descent uses a small subset of gradients instead of a single i_k for more robust outcomes. For $F : \mathbb{R}^d \to \mathbb{R}$ given by

$$F(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x),$$

the gradient

$$\operatorname{grad} F(x) = \frac{1}{n} \sum_{i=1}^{n} \operatorname{grad} f_i(x)$$

is replaced in each iteration by a smaller subset (batch) of gradients over f_i. This means at iteration k, we choose a batch

$$B_k \subseteq \{1, \dots, n\}$$

and use the iteration (update) rule

$$x_{k+1} = x_k - \eta \sum_{i \in B_k} \nabla f_i(x_k).$$

C R T A We started from a finite-dimensional normed vector space described in coordinate-free form. By completion, this space extended to a Banach space and, in the presence of an inner product, to a Hilbert space. The structure gained in this process enabled differential calculus in the form of Fréchet derivatives defined without reference to coordinates.
For computation, coordinates were introduced, enabling partial derivatives. Truncation then replaced the full Taylor expansion by a polynomial of finite degree. Approximation quantified the resulting error via bounds for the Taylor remainder in the underlying norm. Thus, this chapter followed the entire computational cycle: from finite-dimensional coordinate-free structure through completion to infinite-dimensional analysis, and back through coordinates and truncation to finite representation together with controlled approximation.

Essentials

Taylor's formula extends to multivariate functions, providing local polynomial approximations. Local extrema occur at critical points and definiteness of the Hessian matrix enables us to distinguish minima, maxima, and saddle points. Convexity is characterized by positive semi-definiteness of the Hessian. Gradient and stochastic gradient descent are fixed point iterations to numerically determine minimizers.

T A Truncating the Taylor expansion provides low order function representations. The approximation error is controlled by the order of the Taylor polynomial.

Taylor expansion, Taylor polynomial, local extrema, second-order conditions, saddle points, convex function, gradient descent, stochastic gradient descent

T A Multivariate analysis provides local linear models of nonlinear maps through Fréchet derivatives, Jacobians, and Taylor expansions. Once such linearizations are available, the central question is how perturbations of the input influence the computed value. This sensitivity is quantified by the concept of conditioning and forms the starting point of numerical analysis and finite-precision computation.

11.3 Newton's method in $\mathbb{R}^n$

To formulate a fixed point iteration to determine zeros of multivariate functions $f : \mathbb{R}^n \to \mathbb{R}^n$, we need some preparation.

Theorem 11.3.1 (Implicit Function Theorem). *Let $U_1 \subset \mathbb{R}^n$ and $U_2 \subset \mathbb{R}^m$ be open sets, and let*

$$f : U_1 \times U_2 \to \mathbb{R}^m, \qquad (x, y) \mapsto f(x, y),$$

be continuously differentiable. Suppose that

$$f(a, b) = 0$$

for some $(a,b) \in U_1 \times U_2$, and that the Jacobian matrix

$$\frac{\partial f}{\partial y}(a,b) = \begin{pmatrix} \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial y_1} & \cdots & \frac{\partial f_m}{\partial y_m} \end{pmatrix}$$

is invertible.
Then there exist neighborhoods $V_1 \subset U_1$ of a and $V_2 \subset U_2$ of b, together with a continuously differentiable map

$$g : V_1 \to V_2,$$

such that

(i) *$f(x, g(x)) = 0$ for all $x \in V_1$,*

(ii) *Moreover, for all $(x,y) \in V_1 \times V_2$,*

$$f(x,y) = 0 \quad \Longleftrightarrow \quad y = g(x).$$

Theorem 11.3.2 (Inverse Function Theorem)**.** *Let $U \subset \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}^n$ be continuously differentiable. Suppose that $a \in U$ and that $f'(a)$ is invertible.*
Then there exist neighborhoods $U_0 \subset U$ of a and $V_0 \subset \mathbb{R}^n$ of $b = f(a)$ such that $f : U_0 \to V_0$ is bijective and $f^{-1} : V_0 \to U_0$ is continuously differentiable with

$$(f^{-1})'(b) = (f'(a))^{-1}.$$

Proof. Define $F : \mathbb{R}^n \times U \to \mathbb{R}^n$ according to

$$F(x,y) = x - f(y).$$

Then $F(f(a), a) = 0$ and $\frac{\partial F}{\partial y}(x,y) = -f'(y)$. Since $Df(a)$ is invertible, so is $\frac{\partial F}{\partial y}(f(a), a)$.
By the Implicit Function Theorem, there exist neighborhoods

$$V_1 \ni f(a), \qquad V_2 \ni a,$$

and a continuously differentiable map $g : V_1 \to V_2$ such that (according to Part (i))

$$F(x, g(x)) = 0.$$

Hence $x - f(g(x)) = 0$, so that $f(g(x)) = x$ for all $x \in V_1$.
We use Part (ii) to see that $(x,y) \in V_1 \times V_2$ with $F(x,y) = 0$ implies $y = g(x)$. Since then $x = f(y)$, we obtain $y = g(f(y))$. Hence, g is the inverse of f on suitable neighborhoods contained in V_1 and V_2. $\square$

Lemma 11.3.3. *Suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable and $B_r \subseteq \mathbb{R}^n$ a closed ball. Then f is L-Lipschitz continuous on B_r with Lipschitz constant*

$$L = \sup_{z \in B_r} \|f'(z)\|$$

Proof. For $x, y \in B_r$, define $g : [0,1] \to \mathbb{R}^m$ by $g(t) = f(y + t(x-y))$. The chain rule yields

$$g'(t) = f'(y + t(x-y))(x-y).$$

The fundamental theorem of calculus implies

$$f(x) - f(y) = g(1) - g(0) = \int_0^1 f'(y + t(x-y))(x-y)dt.$$

Taking norms on both sides leads to

$$\|f(x) - f(y)\| \leq \int_0^1 \|f'(y + t(x-y))\| \|(x-y)\| dt \leq L\|x-y\|.$$

□

We formulate Newton's method on $\mathbb{R}^n$.

Theorem 11.3.4. *Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ is twice continuously differentiable and $x^* \in \mathbb{R}^n$ such that $f(x^*) = 0$ and $f'(x^*)$ is invertible. Then there is $r > 0$ such that for all $x_0 \in B_r(x^*)$ the iterative sequence*

$$x_{k+1} = x_k - f'(x_k)^{-1} f(x_k) \tag{11.3}$$

converges towards x^* and there is $C > 0$ such that

$$\|x_{k+1} - x^*\| \leq C\|x_k - x^*\|^2.$$

Proof. The inverse function theorem (11.3.2) implies that there is a radius $r > 0$ and a constant $M > 0$ such that

$$\|f'(x)^{-1}\| \leq M, \qquad \forall\, x \in B_r(x^*).$$

Put $e_k := x_k - x^*$ and again observe

$$\int_0^1 f'(x^* + te_k)e_k dt = f(x_k) - f(x^*) = f(x_k).$$

We may add and subtract $f'(x_k)e_k$ to obtain

$$f(x_k) = f'(x_k)e_k + \int_0^1 (f'(x^* + te_k - f'(x_k))e_k dt.$$

By subtracting x^* from both sides of Newton's iteration, we get

$$\begin{aligned} e_{k+1} &= e_k - f'(x_k)^{-1}\Big(f'(x_k)e_k + \int_0^1 (f'(x^* + te_k) - f'(x_k))e_k dt\Big) \\ &= -f'(x_k)^{-1} \int_0^1 (f'(x^* + te_k) - f'(x_k))e_k dt. \end{aligned}$$

We may take norms on both sides and the triangle inequality still holds for the integration (as a limit of summations),

$$\begin{aligned}
\|e_{k+1}\| &\leq \|f'(x_k)^{-1}\| \int_0^1 \|f'(x^* + te_k) - f'(x_k)\| \|e_k\| dt \\
&\leq \|f'(x_k)^{-1}\| \int_0^1 L\|x^* + te_k - x_k\| \|e_k\| dt \\
&= \|f'(x_k)^{-1}\| L \int_0^1 \|(t-1)e_k\| \|e_k\| dt \\
&= \|f'(x_k)^{-1}\| L \|e_k\|^2 \int_0^1 (1-t) dt \\
&= \|f'(x_k)^{-1}\| L \|e_k\|^2 \frac{1}{2} \\
&\leq \frac{ML}{2} \|e_k\|^2.
\end{aligned}$$

This is the quadratic error estimate with $C := \frac{ML}{2}$. To check for convergence, we decrease the radius of the ball centered at x^*. Choose a radius $\rho > 0$ such that $\rho \leq r$ and $C\rho \leq \theta < 1$. For $x_0 \in B_\rho(x^*)$, we have $\|e_0\| \leq \rho < 1$ and induction yields $\|e_k\| \leq \rho$, so that the chain of inequalities leads to

$$\|e_{k+1}\| \leq C\|e_k\|^2 \leq C\rho\|e_k\| = \theta\|e_k\| \leq \dots \theta^{k+1}\|e_0\| \to 0.$$

Thus, $x_k \to x^*$. □

To compute $f'(x_k)^{-1} f(x_k)$, we usually do NOT invert the matrix $f'(x_k)$ but instead solve the linear system of equations

$$f'(x_k)z = f(x_k)$$

for $z \in \mathbb{R}^n$.

Example 11.3.5. *Consider $f : \mathbb{R}^n \to \mathbb{R}^n$ given by*

$$f(x) = \begin{pmatrix} x_1^2 - a_1 \\ \vdots \\ x_n^2 - a_n \end{pmatrix}, \qquad a_1, \dots, a_n > 0.$$

To find $x^* \in \mathbb{R}^n$ such that $f(x^*) = 0$, we first determine the Jacobian

$$f'(x) = \operatorname{diag}(2x_1, \dots, 2x_n)$$

As opposed to the numerical implementation, we here determine the inverse matrix

$$f'(x)^{-1} = \operatorname{diag}(\frac{1}{2x_1}, \dots, \frac{1}{2x_n}).$$

The iteration is

$$\begin{aligned}
x^{(k+1)} &= x^{(k)} - \operatorname{diag}(\frac{1}{2x_1}, \dots, \frac{1}{2x_n}) f(x^{(k)}) \\
&= x^{(k)} - \begin{pmatrix} \frac{(x_1^{(k)})^2 - a_1}{2x_1^{(k)}} \\ \vdots \\ \frac{(x_n^{(k)})^2 - a_n}{2x_n^{(k)}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(x_1^{(k)} + \frac{a_1}{x_1^{(k)}}) \\ \vdots \\ \frac{1}{2}(x_n^{(k)} + \frac{a_n}{x_n^{(k)}}) \end{pmatrix}.
\end{aligned}$$

The iteration splits into each separate component. This is a familiar iteration in one dimension.

Example 11.3.6. *Consider $g(y) = y^2 - b$ for $b > 0$. The Newton iteration for the 1-dimensional function is*

$$y^{(k+1)} = y^{(k)} - \frac{(y^{(k)})^2 - b}{2y^{(k)}} = \frac{1}{2}(y^{(k)} + \frac{b}{y^{(k)}}),$$

and $y^{(k)} \to \sqrt{b}$ for suitable choice of $y^{(0)}$.

Thus, for suitable choices of $x^{(0)} \in \mathbb{R}^n$, we have $x^{(k)} \to \begin{pmatrix} \sqrt{a_1} \\ \vdots \\ \sqrt{a_n} \end{pmatrix}$.

The Newton iteration is a fixed point iteration by

$$\Phi(x) = x - f'(x)^{-1} f(x),$$

since $x_{k+1} = \Phi(x_k)$ becomes (11.3) with

$$f(x^*) = 0 \quad \Leftrightarrow \quad \Phi(x^*) = x^*.$$

Since $f(x^*) = 0$, the derivative of Φ is

$$\Phi'(x^*) = I_n - \big(f'(\cdot)f(\cdot)\big)'(x^*) = I_n - f'(x^*)^{-1} f'(x^*) = I_n - I_n = 0.$$

Thus, we have $\|\Phi'(x^*)\| = 0$. Banach-fixed-point theorem only needs $\|\Phi'(x^*)\| < 1$ to establish linear convergence, cf. Theorem 10.4.12. The Newton iteration leads to $\|\Phi'(x^*)\| = 0$ and then derives quadratic convergence.

Example 11.3.7. *Fix an integer $m \geq 2$. By identifying $\mathbb{C}$ with $\mathbb{R}^2$, the complex function $\mathbb{C} \to \mathbb{C}$, $z \mapsto z^m - 1$, is written as $f : \mathbb{R}^2 \to \mathbb{R}^2$,*

$$f(x, y) = \begin{pmatrix} \mathrm{Re}((x+iy)^m - 1) \\ \mathrm{Im}((x+iy)^m - 1) \end{pmatrix}.$$

Its zeros are the m-th roots of unity $\omega_j = e^{2\pi i j/m}$, $j = 0, \ldots, m-1$.
To determine the Newton iteration in terms of z, we need to determine the Jacobian of f denoted by $f'(z_n) = f'(x_n, y_n)$. Let $h \in \mathbb{C}$ be a small increment. The binomial theorem,

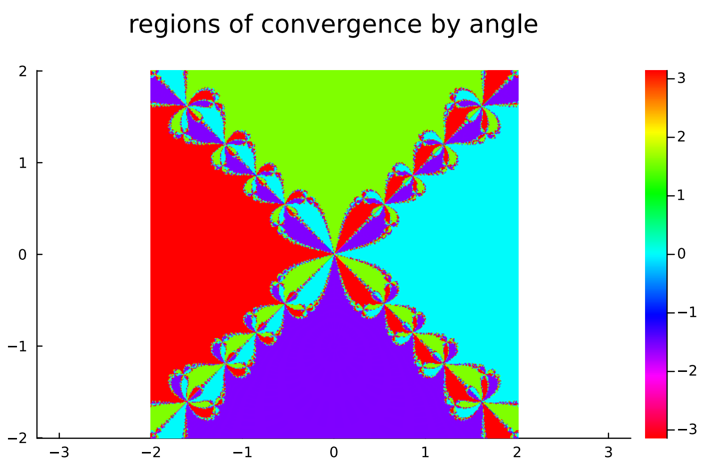
$$(z+h)^m = z^m + m z^{m-1} h + \sum_{j=2}^{m} \binom{m}{j} z^{m-j} h^j,$$

implies that $f'(z)h = m z^{m-1} h$. The inverse is multiplication by $\frac{1}{m z^{m-1}}$, so that the Newton iteration becomes

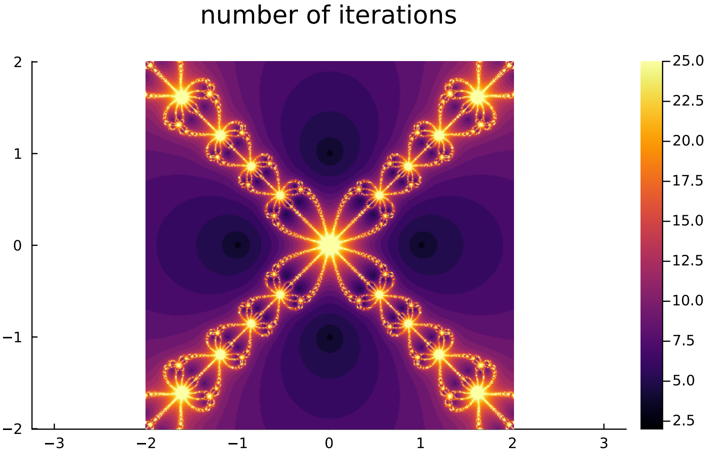
$$z_{k+1} = z_k - \frac{f(z_k)}{f'(z_k)} = z_k - \frac{z_k^m - 1}{m z_k^{m-1}}.$$

We investigate numerically for which initializations $z_0 \in \mathbb{C}$ the Newton iterations converges and if so then to which root.

```
function myNewton(f, ∂f, z0; tol=1e-8, maxiter=100)
    z = copy(z0)
    for k in 1:maxiter
        z = z-f(z)/∂f(z)
        if abs(f(z))<tol
            return z, k
        end
    end
    return z, maxiter
end

function newtonFractal(f,∂f,s)
    step = 0.1e-1 # 10^-2
    interval = -s:step:s
    L = length(interval)

    Zgrid = [a+b*im for b in interval, a in interval]

    Y = myNewton.(f,∂f,Zgrid;tol=1e-10,maxiter=100)
    Z = [Y[k,l][1] for k in 1:L, l in 1:L]
    K = [Y[k,l][2] for k in 1:L, l in 1:L]

    return Z, K, interval
end

m = 4
f(z) = z^m-1
fprime(x) = m*x^(m-1)

X, K, interval = newtonFractal(f,fprime,2);

myColor = range(HSV(0,1,1), stop = HSV(-360,1,1), length = 90)
heatmap(interval, interval, angle.(X), color = myColor, clim = (-π,π), aspect_ratio
    =:1, title = "regions of convergence by angle")
```




```
heatmap(interval,interval,K, clim = (2,25),  aspect_ratio=:equal,title = "number of
    iterations")
```



Where on the real axis is no convergence? The Newton iteration breaks when there is some $z_k = 0$. Obviously, there cannot be any convergence for $z_0 = 0$.

Assume $m = 3$. The Newton iteration then is

$$z_{k+1} = z_k - \frac{z_z^3 - 1}{3z_k^2} = \frac{2}{3}z_k + \frac{1}{3z_k^2} \tag{11.4}$$

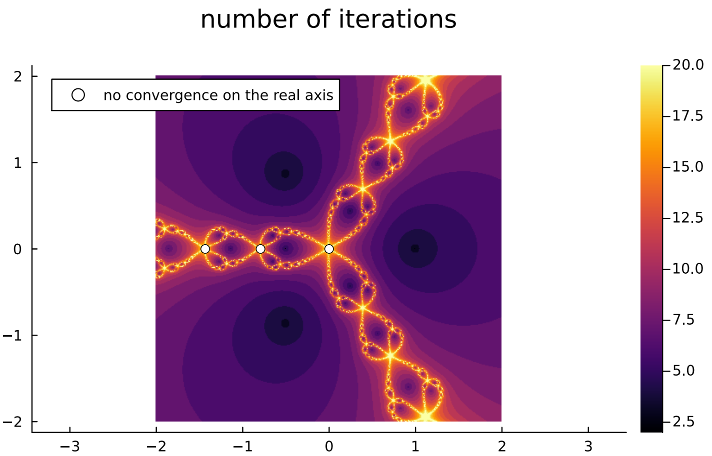
If $z_1 = 0$, then we have

$$0 = \frac{2}{3}z_0 + \frac{1}{3z_0^2} \qquad \Rightarrow \qquad z_0 = -\left(\frac{1}{2}\right)^{1/3}. \tag{11.5}$$

Next, we consider $z_2 = 0$. This yields $z_1 = -\left(\frac{1}{2}\right)^{1/3}$, which leads to

$$-\left(\frac{1}{2}\right)^{1/3} = \frac{2}{3}z_0 + \frac{1}{3z_0^2} \qquad \Rightarrow \qquad \cdots \qquad \Rightarrow \qquad z_0 \approx -1.4338 \tag{11.6}$$

In that case, we obtain:



Essentials

The inverse function theorem is a consequence of the implicit function theorem and ensures that an invertible Jacobi matrix implies local invertibility. It guarantees that Newton's method in d dimensions converges locally, which solves nonlinear systems of equations by a fixed point iteration.

A The Newton iterations approximate solutions to nonlinear systems of equations.

implicit function theorem, inverse function theorem, Newton iteration

11.4 Local extrema under constraints

Minimization of a function $f : \mathbb{R}^n \to \mathbb{R}$ is often constraint to complicated subsets of $\mathbb{R}^n$. We derive the tools to compute local extrema under constraints.

Definition 11.4.1 (submanifolds)**.** *A subset $M \subset \mathbb{R}^n$ is called a k-dimensional submanifold if for every $x_0 \in M$, there exist*

- *an open neighborhood $U \subset \mathbb{R}^n$ of x_0,*
- *a continuously differentiable function $g : U \to \mathbb{R}^r$ with $r = n - k$,*

such that

$$M \cap U = \{x \in U : g(x) = 0\}, \qquad \operatorname{rank} g'(x) = r, \quad \forall x \in M \cap U.$$

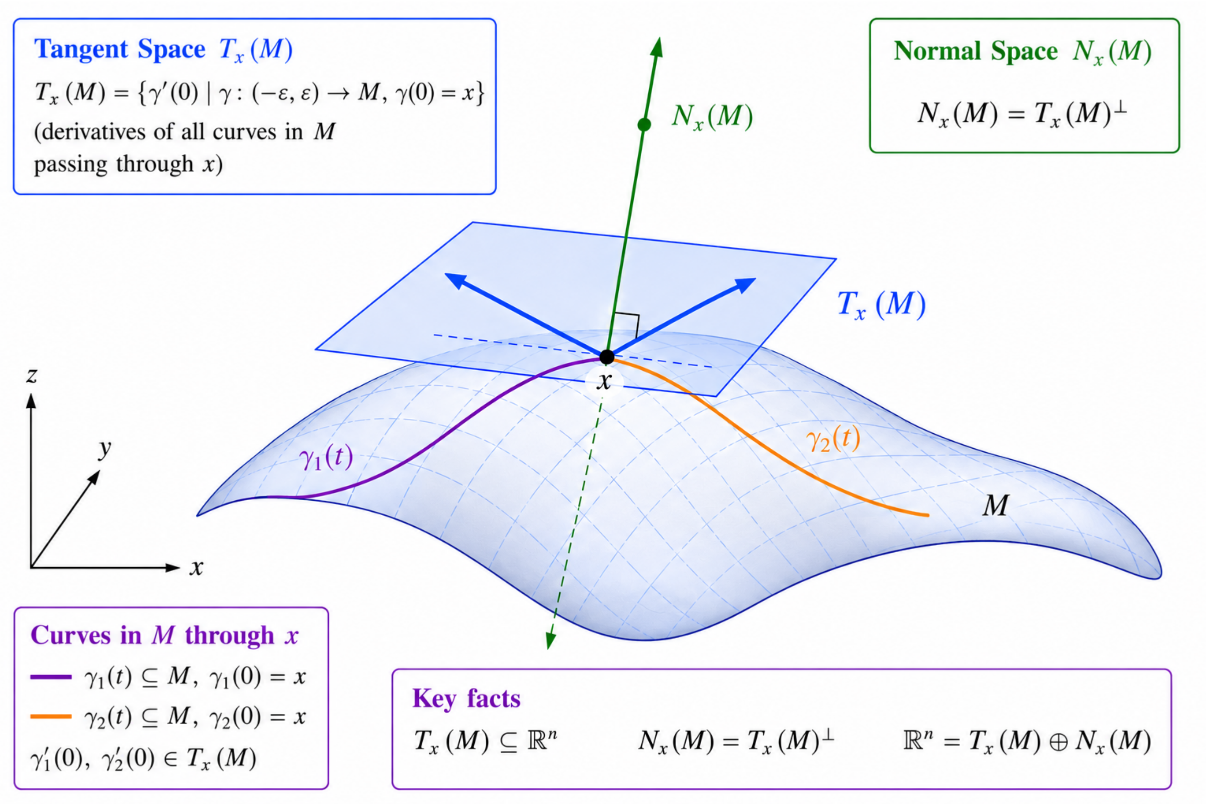
Example 11.4.2. *Consider*

$$M = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 - 1 = 0\}.$$

Then M is a 1-dimensional submanifold of $\mathbb{R}^2$.
Indeed, for $U = \mathbb{R}^2$, the function $g : \mathbb{R}^2 \to \mathbb{R}$ given by

$$g(x) = x_1^2 + x_2^2 - 1,$$

yields $g'(x) = (2x_1, 2x_2)$, which has rank one for $x \in M \cap U = M$ since $(0,0)^\top \notin M$.



Definition 11.4.3 (tangent space). *Let $M \subset \mathbb{R}^n$ be a k-dimensional submanifold and let $x \in M$.*

(a) The tangent space of M at x is

$$T_x(M) = \left\{\gamma'(0) : \gamma : (-\varepsilon,\varepsilon) \to M \text{ continuously differentiable},\ \gamma(0) = x\right\}.$$

(b) The normal space of M at x is

$$N_x(M) = T_x(M)^\perp.$$

By definition, the normal space $N_x(M)$ is a linear subspaces of $\mathbb{R}^n$.

Definition 11.4.4. *Let $\Omega \subseteq \mathbb{R}^n$ be open and $f : \Omega \to \mathbb{R}$. Suppose $M \subseteq \mathbb{R}^n$ is a k-dimensional submanifold. We say that $x \in M \cap \Omega$ is a local maximum of f on M if there is an open neighborhood $U \subseteq \mathbb{R}^n$ of x such that*

$$f(y) \le f(x), \qquad \forall y \in U \cap M \cap \Omega.$$

A local minimum of f on M is defined analogously.

Theorem 11.4.5 (extrema under constraints). *Let $\Omega \subset \mathbb{R}^n$ be open, $f : \Omega \to \mathbb{R}$ be continuously differentiable, and suppose that M is a k-dimensional submanifold. If $x_0 \in \Omega \cap M$ is* *a local extremum of f on M, then*

$$\nabla f(x_0) \in N_{x_0}(M).$$

Proof. Let $\gamma : (-\varepsilon, \varepsilon) \to M$ be a continuously differentiable curve such that $\gamma(0) = x_0$. Define $F : (-\varepsilon, \varepsilon) \to \mathbb{R}$ by $F(t) = f(\gamma(t))$. Since f has a local extremum at x_0 on M, the function F has a local extremum at 0 and hence $F'(0) = 0$. The chain rule implies

$$0 = F'(0) = f'(x_0)\gamma'(0) = \langle \nabla f(x_0), \gamma'(0) \rangle.$$

Since $\gamma'(0) \in T_{x_0}(M)$ was arbitrary,

$$\nabla f(x_0) \perp T_{x_0}(M).$$

Therefore, $\nabla f(x_0) \in N_{x_0}(M)$. $\square$

Theorem 11.4.6. *Let $M \subset \mathbb{R}^n$ be a k-dimensional submanifold and $x \in M$ with a neighborhood $U \subseteq \mathbb{R}^n$ of x such that $g : U \to \mathbb{R}^r$ with $r = n - k$ is continuously differentiable with $M \cap U = \{y \in U : g(y) = 0\}$ and $\operatorname{rank} g'(y) = r$ for all $y \in M \cap U$. Then*

$$T_x(M) = \{\nabla g_1(x), \ldots, \nabla g_r(x)\}^\perp,$$
$$N_x(M) = \operatorname{span}\{\nabla g_1(x), \ldots, \nabla g_r(x)\}.$$

Since the Jacobian of g is

$$g'(x) = \begin{pmatrix} \operatorname{grad} g_1(x) \\ \vdots \\ \operatorname{grad} g_r(x) \end{pmatrix} = \begin{pmatrix} (\nabla g_1(x))^\top \\ \vdots \\ (\nabla g_r(x))^\top \end{pmatrix},$$

the rank requirement ensures that $\{\nabla g_1(x), \ldots, \nabla g_r(x)\}$ are linearly independent. The theorem further says that $T_x(M) = \operatorname{null} g'(x) = \{v \in \mathbb{R}^n : g'(x)v = 0\}$ is a linear subspace of dimension k, $N_x(M)$ is a $n - k$-dimensional linear subspace, and

$$\mathbb{R}^n = T_x(M) \oplus N_x(M).$$

Proof. To show $T_x(M) \subseteq \{\nabla g_1(x), \ldots, \nabla g_r(x)\}^\perp$, let $v \in T_x(M)$. By definition of the tangent space, there exists a continuously differentiable curve $\gamma : (-\varepsilon, \varepsilon) \to M$ such that $\gamma(0) = x$ and $\gamma'(0) = v$. Since $\gamma(t) \in M$ for all $t \in (-\varepsilon, \varepsilon)$, we have

$$g_i(\gamma(t)) = 0, \qquad i = 1, \ldots, r.$$

Differentiating and applying the chain rule yields

$$0 = \frac{d}{dt} g_i(\gamma(t))\Big|_{t=0} = g_i'(x)\gamma'(0) = g_i'(x)v = \langle \nabla g_i(x), v \rangle, \qquad i = 1, \ldots, r.$$

Thus, we have verified $T_x(M) \subseteq \{\nabla g_1(x), \ldots, \nabla g_r(x)\}^\perp$.
To verify the reverse inclusion, take $v \in \mathbb{R}^n$ such that $v \perp \nabla g_i(x) = 0$, for $i = 1, \ldots, r$. Define $F : \mathbb{R} \times \mathbb{R}^r \to \mathbb{R}^r$ by

$$F(t, y) = g(x + tv + (g'(x))^\top y).$$

Then we have $F(0,0) = g(x) = 0$ and $\frac{\partial F}{\partial y}(0,0) = g'(x)(g'(x))^\top$. Since $g'(x)(g'(x))^\top \in \mathbb{R}^{r\times r}$ is invertible, the implicit function theorem implies that there is $\varepsilon > 0$ and a continuously differentiable $\eta : (-\varepsilon, \varepsilon) \to \mathbb{R}^r$ such that $\eta(0) = 0$ and

$$F(t, \eta(t)) = 0.$$

Thus, the curve $\gamma : (-\varepsilon, \varepsilon) \to M$,

$$\gamma(t) = x + tv + (g'(x))^\top \eta(t)$$

is continuously differentiable and indeed maps into M because

$$g(\gamma(t)) = F(t, \eta(t)) = 0.$$

Differentiating at $t = 0$ yields

$$0 = g'(x)\gamma'(0) = g'(x)(v + (g'(x))^\top \eta'(0)) = g'(x)v + g'(x)(g'(x))^\top \eta'(0).$$

The assumption on v implies $g'(x)v = 0$, so that we derive

$$g'(x)(g'(x))^\top \eta'(0) = 0.$$

Since $g'(x)(g'(x))^\top$ is invertible, we must have $\eta'(0) = 0$. Therefore

$$\gamma'(0) = v + (g'(x))^\top \eta'(0) = v.$$

Thus $v \in T_x(M)$, and we have verified that $\{\nabla g_1(x), \dots, \nabla g_r(x)\}^\perp \subseteq T_x(M)$. □

Corollary 11.4.7. *Let $U \subset \mathbb{R}^n$ be open and let*

$$M = \{x \in U : g_1(x) = \cdots = g_r(x) = 0\}$$

be a k-dimensional submanifold and $\operatorname{rank} g'(y) = r$ *for all $y \in M \cap U$. Let*

$$f : U \to \mathbb{R}$$

be continuously differentiable and suppose that $f|_M$ has a local extremum at $x_0 \in M$. Then there are coefficients $\lambda_1, \dots, \lambda_r \in \mathbb{R}$ such that

$$\nabla f(x_0) = \sum_{i=1}^{r} \lambda_i \nabla g_i(x_0).$$

Example 11.4.8. *Find the extrema of $f(x) = x_1 x_2$ subject to the constraint $g(x) = x_1^2 + x_2^2 - 1 = 0$.*
We compute

$$\nabla f(x) = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}, \qquad \nabla g(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}.$$

At a constrained extremum, $\nabla f(x) = \lambda \nabla g(x)$, so

$$
\begin{aligned}
x_2 &= 2\lambda x_1, \\
x_1 &= 2\lambda x_2, \\
x_1^2 + x_2^2 - 1 &= 0.
\end{aligned}
$$

Multiplying the first two equations by x_1 and x_2, respectively makes the right hand sides equal and we derive $x_1^2 = x_2^2$, hence, $x_1 = \pm x_2$.
Case 1: $x_1 = x_2$
Using the third equations, we obtain $2x_1^2 = 1$, thus $x_1 = x_2 = \pm\frac{1}{\sqrt{2}}$. The function value is $f(x) = \frac{1}{2}$.
Case 2: $x_1 = -x_2$
Again, $2x_1^2 = 1$, so that $x_1 = \pm\frac{1}{\sqrt{2}}$ and $x_2 = \mp\frac{1}{\sqrt{2}}$. The function value is $f(x) = -\frac{1}{2}$.
Since the circle $\{x \in \mathbb{R}^2 : g(x) = 0\}$ is compact and f is continuous, it must attain its maximum and its minimum. Those are $1/2$ ad $= -1/2$, respectively.

Example 11.4.9. *Consider $f : \mathbb{R}^3 \to \mathbb{R}$ given by $f(x) = \|x\|^2$ and find the minimizer on $M = \{x \in \mathbb{R}^3 : g(x) = 0\}$, where*

$$
\begin{aligned}
g_1(x) &= x_1^2 + x_2^2 - 1 = 0 \\
g_2(x) &= x_3 - x_1 = 0.
\end{aligned}
$$

The Jacobian of g is

$$
g(x) = \begin{pmatrix} 2x_1 & 2x_2 & 0 \\ -1 & 0 & 1 \end{pmatrix},
$$

and has rank 2 for all $x \in M$. By Corollary 11.4.7, any constrained extremum satisfies $\nabla f(x) = \lambda_1 \nabla g_1(x) + \lambda_2 \nabla g_2(x)$. Since

$$
\nabla f = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 2x_3 \end{pmatrix}, \qquad \nabla g_1 = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 0 \end{pmatrix}, \qquad \nabla g_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},
$$

we obtain

$$
\begin{aligned}
2x_1 &= 2\lambda_1 x_1 - \lambda_2 \\
2x_2 &= 2\lambda_1 x_2 \\
2x_3 &= \lambda_2.
\end{aligned}
$$

Using the constraint $x_3 = x_1$, the last equation gives $\lambda_2 = 2x_1$. Substituting into the first equation yields $(2 - \lambda_1)x_1 = 0$, while the second equation becomes $(1 - \lambda_1)x_2 = 0$. If $x_1 = 0$, then $x_2^2 = 1$ and therefore $x = (0, \pm 1, 0)^\top$ with $\lambda = (1, 0)^\top$.
If $x_1 \neq 0$, then $\lambda_1 = 2$, so that $x_2 = 0$, and hence $x_1^2 = 1$ and $x = (\pm 1, 0, \pm 1)^\top$ with $\lambda = (2, \pm 2)^\top$.
Evaluating the objective,

$$
f(0, \pm 1, 0) = 1, \qquad f(\pm 1, 0, \pm 1) = 2.
$$

Therefore the constrained minimizers are $(0, \pm 1, 0)^\top$ with minimal value $f(0, \pm 1, 0) = 1$.

Newton's method for constrained extrema

Let

$$f : \mathbb{R}^n \to \mathbb{R}$$

and let

$$M = \{x \in \mathbb{R}^n : g(x) = 0\}, \qquad g : \mathbb{R}^n \to \mathbb{R}^r,$$

be a k-dimensional submanifold. We define the Lagrangian by

$$L : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}, \qquad L(x, \lambda) = f(x) - \lambda^\top g(x).$$

To identify local extrema, we must find zeros of the function

$$F : \mathbb{R}^{n+r} \to \mathbb{R}^{n+r}, \qquad F(x, \lambda) = \begin{pmatrix} \nabla_x L(x, \lambda) \\ g(x) \end{pmatrix} = \begin{pmatrix} \nabla f(x) - \sum_{i=1}^r \lambda_i \nabla g_i(x) \\ g(x) \end{pmatrix}.$$

Newton's method applied to F yields the iteration

$$(x_{m+1}, \lambda_{m+1}) = (x_m, \lambda_m) - F'(x_m, \lambda_m)^{-1} F(x_m, \lambda_m),$$

provided that the Jacobian matrix $F'(x_m, \lambda_m)$ is invertible. The derivative of F uses the Hessian of L with respect to x, $H_x L$, and is given by

$$F'(x, \lambda) = \begin{pmatrix} H_x L(x, \lambda) & -J_g(x)^\top \\ J_g(x) & 0 \end{pmatrix}.$$

```
using LinearAlgebra
using ForwardDiff

function myNewtonConstrained(f, g, x0, λ0; tol=1e-10, maxiter=30)
    x = copy(x0)
    n = length(x)
    λ = copy(λ0)
    r = length(λ)
    for k in 1:maxiter
        ∇f = ForwardDiff.gradient(f, x)
        G  = ForwardDiff.jacobian(g, x)
        L(z) = f(z) - dot(λ, g(z))
        H = ForwardDiff.hessian(L, x)
        F = vcat(∇f - G'λ, g(x))
        if norm(F) < tol return x, λ, k end
        K = [H   -G'; G   zeros(r, r)]
        Δ = -K \ F
        x += Δ[1:n]
        λ += Δ[n+1:end]
    end
    return x, λ, maxiter
end

f(x) = x[1] * x[2]
g(x) = [x[1]^2 + x[2]^2 - 1]
x0 = [4.1, 1.4]
λ0 = [0.0]
x, λ, it = myNewtonConstrained(f, g, x0, λ0)
println("iterations = ", it), println("λ = ", λ), println("x = ", x)
println("f(x) = ", f(x)), println("g(x) = ", g(x));
```

```
iterations = 8
λ= [0.4999999999999491]
x = [0.707106781186652, 0.707106781186652]
f(x) = 0.5000000000001478
g(x) = [2.9554136915521667e-13]
```

```
f(x) = x[1]^2+x[2]^2+x[3]^2
g(x) = [x[1]^2 + x[2]^2 - 1; x[3] - x[1]]
x0 = [1, 1, 1]
λ0 = [0.0, 0.0]
x, λ, it = myNewtonConstrained(f, g, x0, λ0)
println("iterations = ", it), println("λ = ", λ), println("x = ", x)
println("f(x) = ", f(x)), println("g(x) = ", g(x));
```

```
iterations = 7
λ= [1.0, 0.0]
x = [0.0, 1.0, 0.0]
f(x) = 1.0
g(x) = [0.0, 0.0]
```